

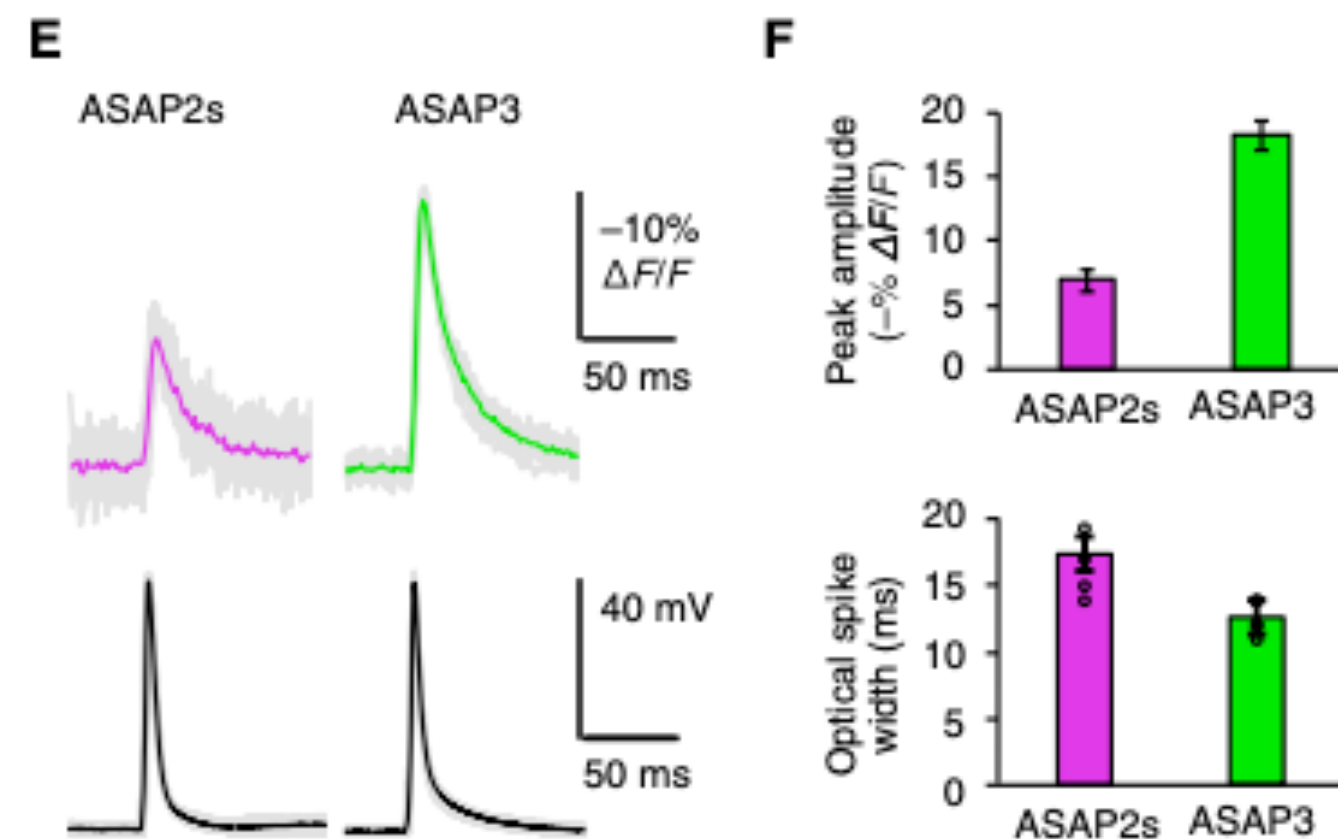
High Speed Imaging

‘Breaking the KHz barrier’

Jacques Carolan

Why

- Calcium dynamics ~ 100 ms vs APs ~ 1 ms
- Genetically Encoded Voltage Indicators:
 - ASAP3¹, QuasAr3², JEDI³



- Sub-threshold events
- All-optical electrophysiology⁴ & connectomics

¹ Villette et al., (2019) [2P]

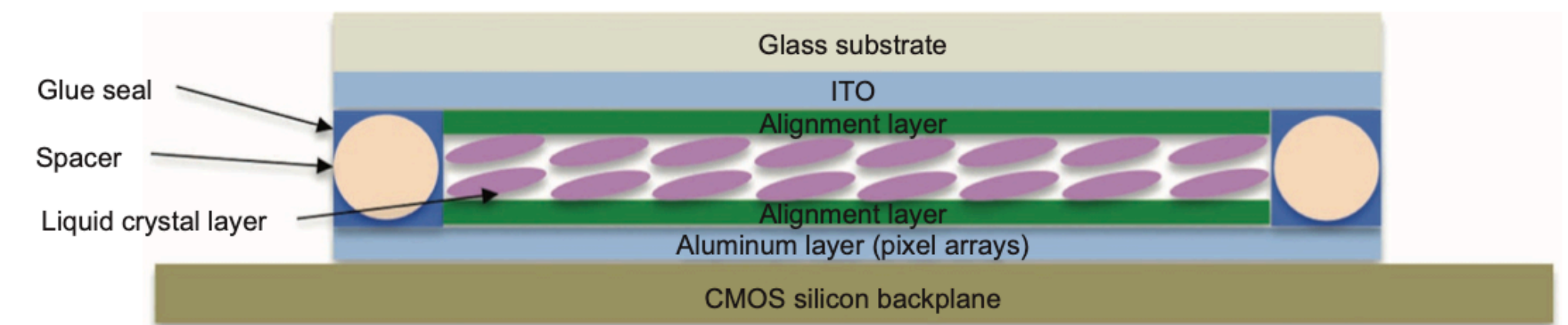
² Adam et al., (2019) [1P]

³ St-Pierre Lab, Zoe + Junlang! [2P]

⁴ Y. Adam (2021)

Challenges

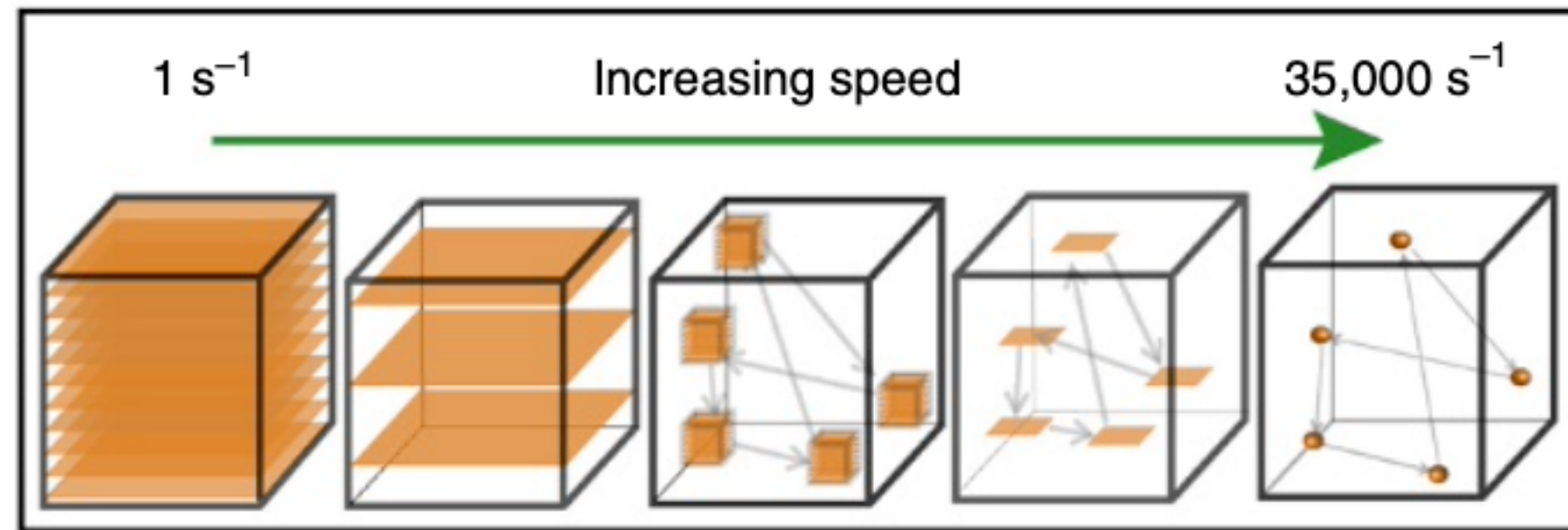
- Beam steering (e.g. Galvo):
 - Moving a mass (e.g. a mirror) has an inherent bandwidth $\rightarrow \sim 10$ KHz = 0.1 ms
- Beam patterning (e.g. SLM):
 - Vicious liquid crystals $\rightarrow \sim 100$ Hz = 10 ms



Zhang et al., Science & Applications (2014)

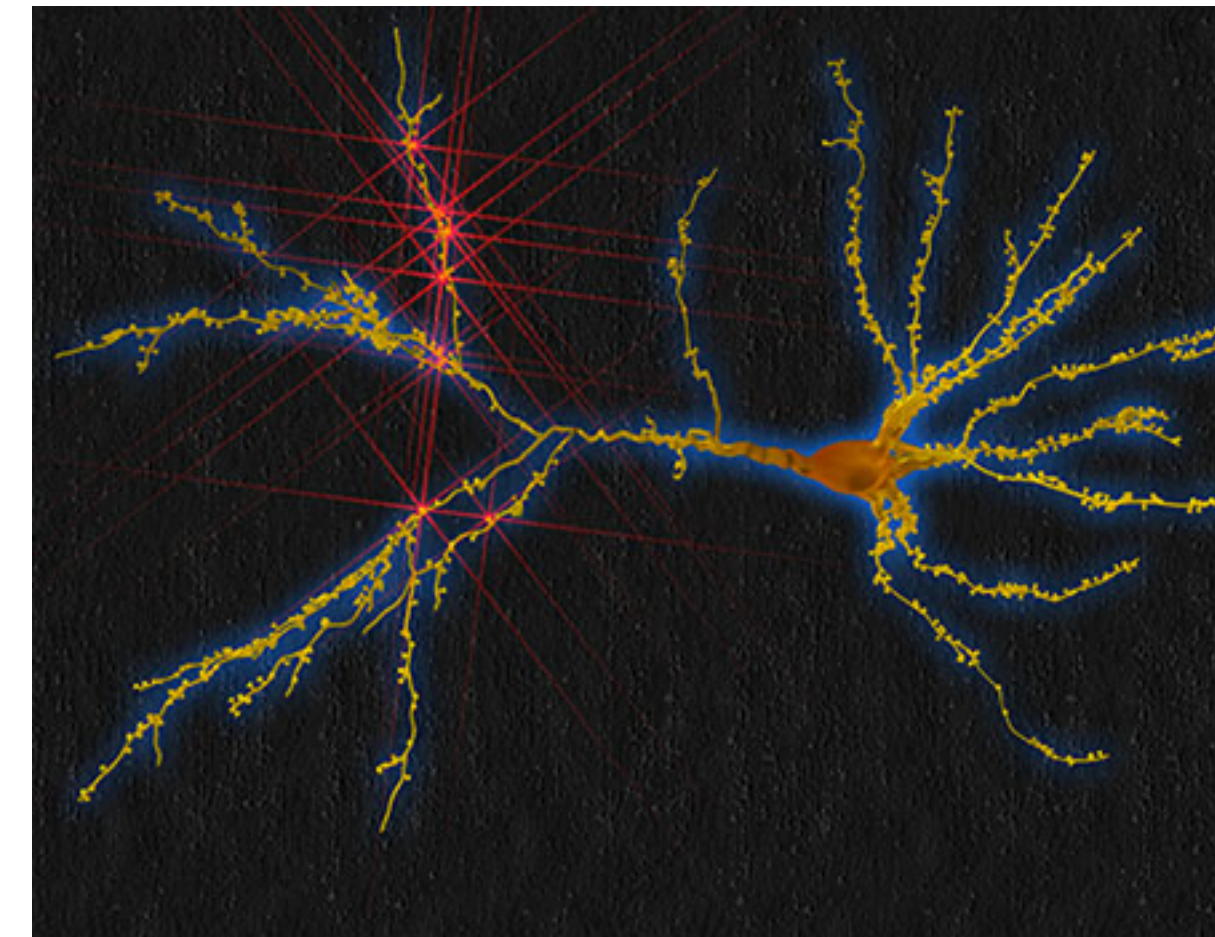
- Cameras (e.g. sCMOS):
 - 100 fps (2000 x 2000 px)

Random Access



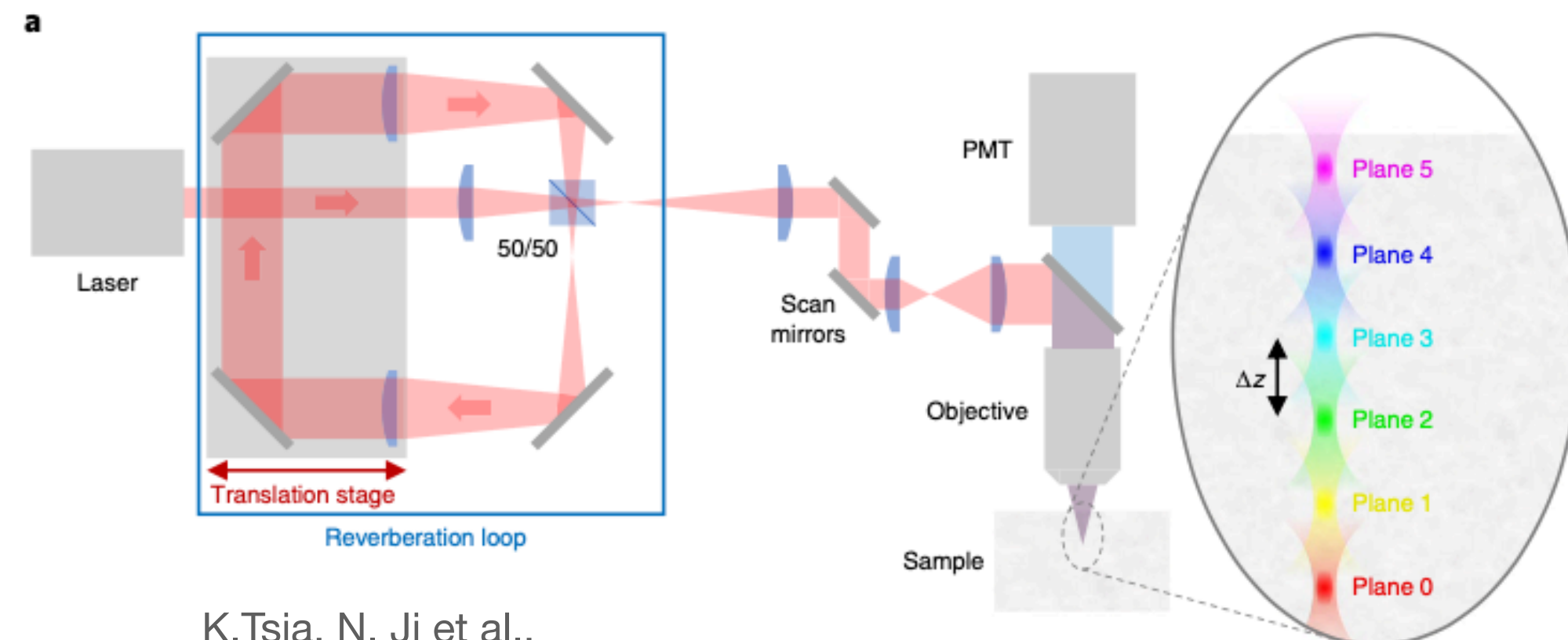
K. M. N. S. Nadella ... P. A. Kirkby, and R. A. Silver, *Nat Meth* (2016).

Computational



K. Podgorski

Multiplexing



K.Tsia, N. Ji et al.,

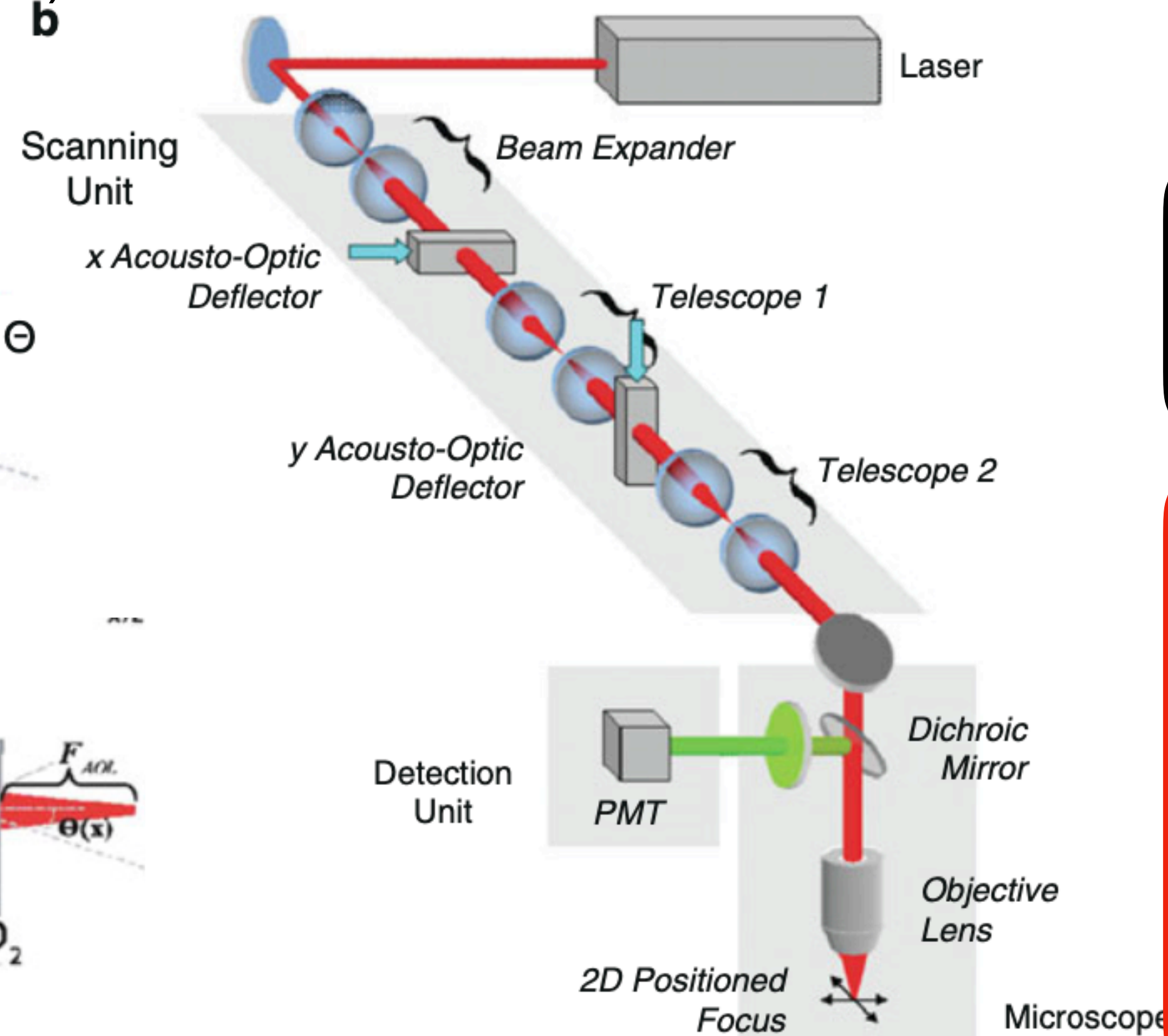
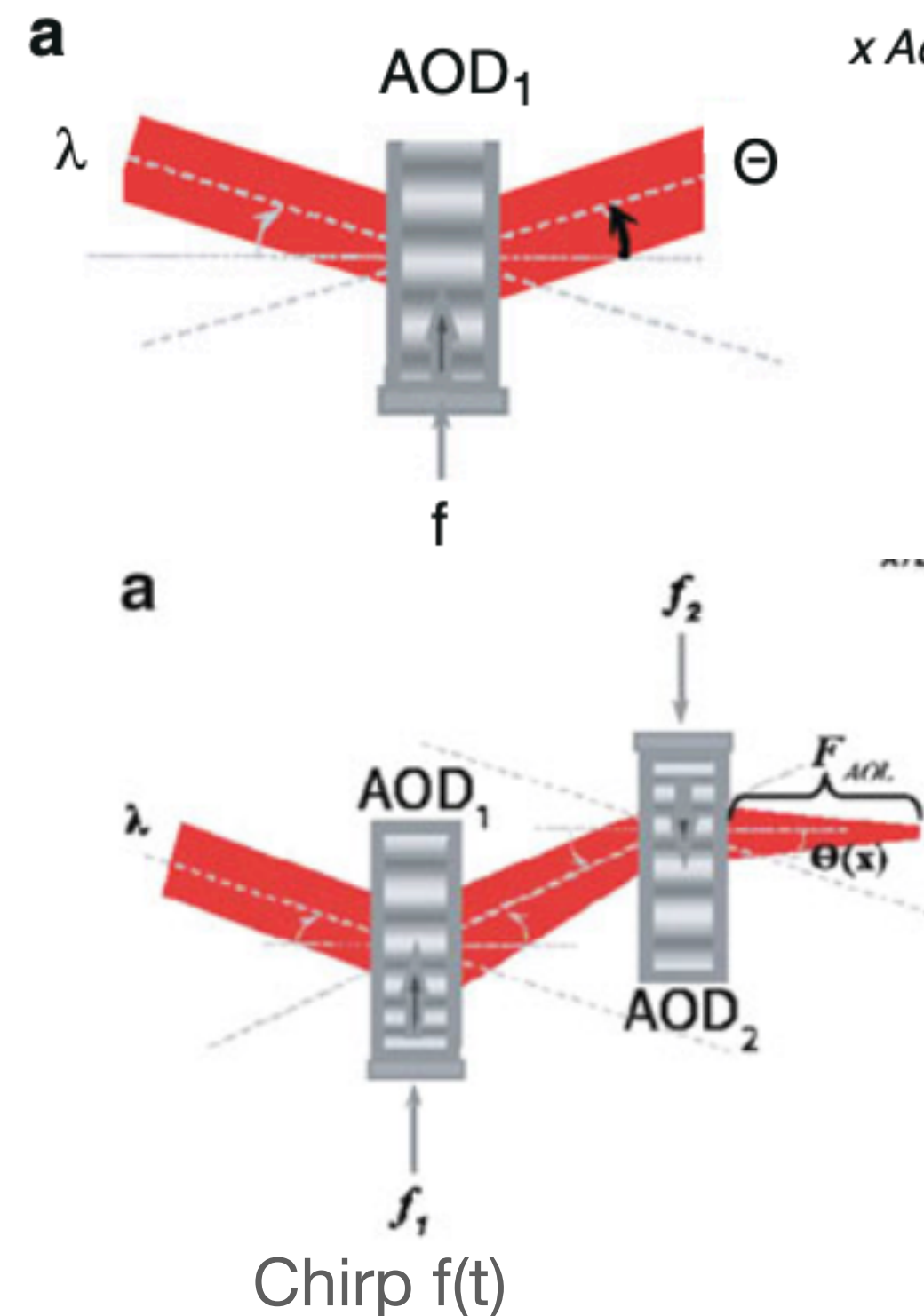
Aims:

- Broad overview of approaches
- Technological advances to keep an eye on
- Opportunities for us

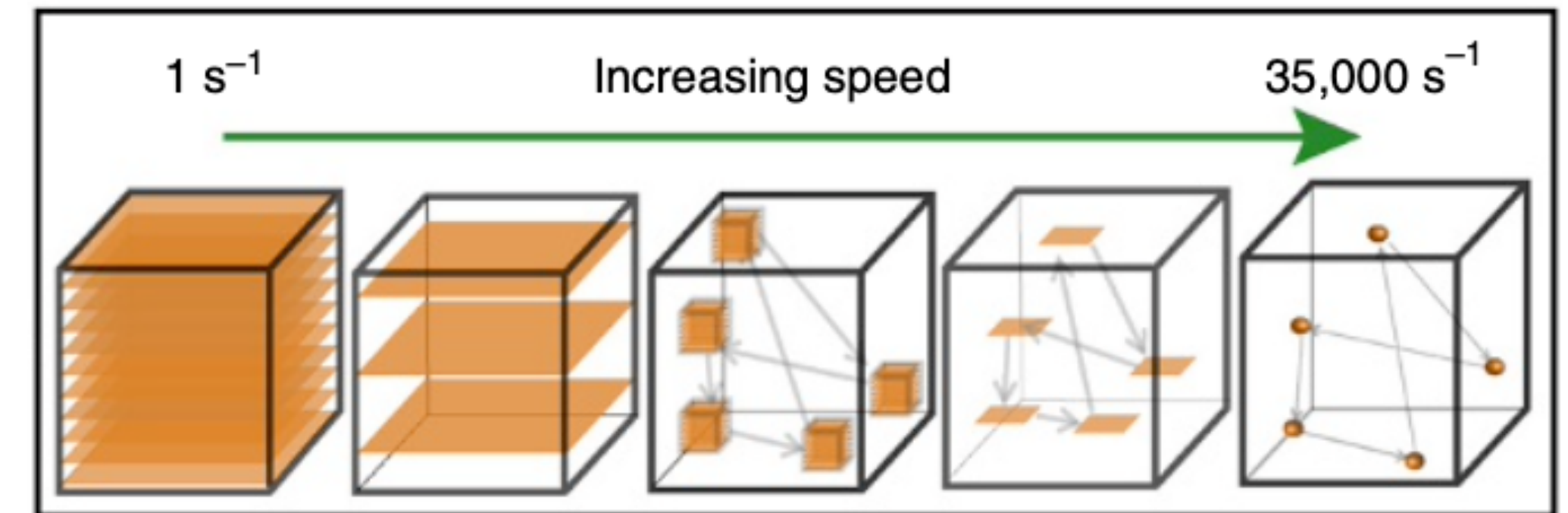
Random Access Microscopy

- Only access predetermined ROI's
- Programmable scanning enabled by **acoustic optic deflectors (AODs)**

$$\theta \approx \frac{\lambda f}{\nu}$$



G. D. Reddy et al., (2015)

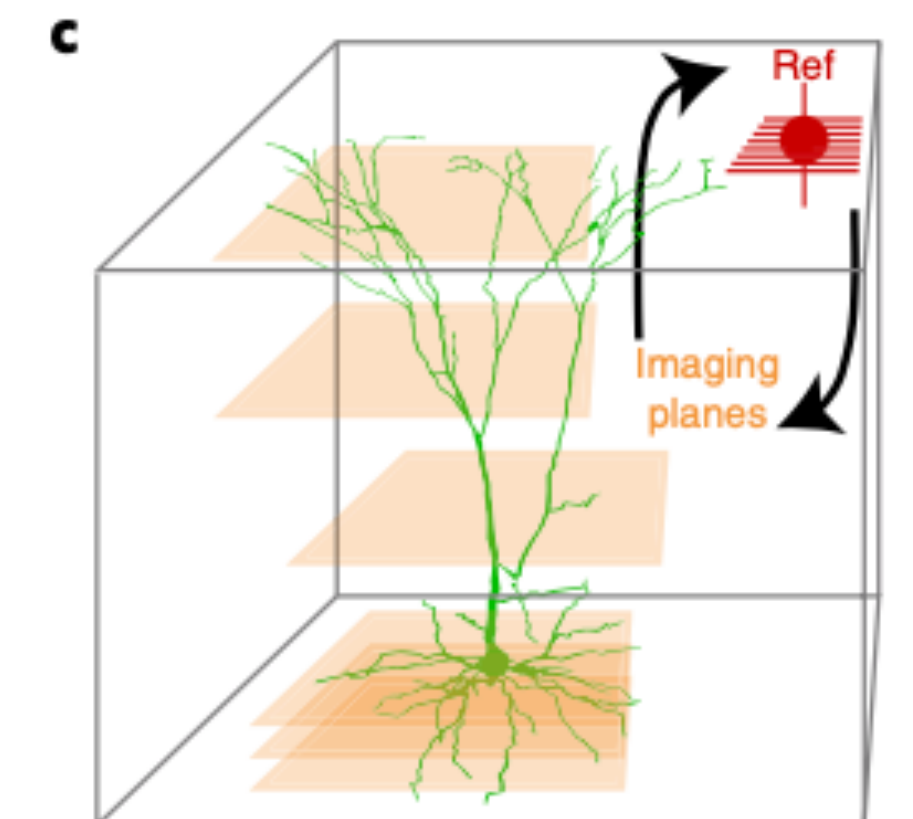


K. M. N. S. Nadella ... P. A. Kirkby, and R. A. Silver, *Nat Meth* (2016).

Materials: Tellerium Dioxide / Quartz / **Glass**
Speed of sound = 4000 m/s
Switching speed = 20 mm / 4000 m/s = **5 μs**

+Fast
 +3D
 +Versatile
 +Well developed tech

- Sparsity
 - Dispersion comp.
 - Movement correction

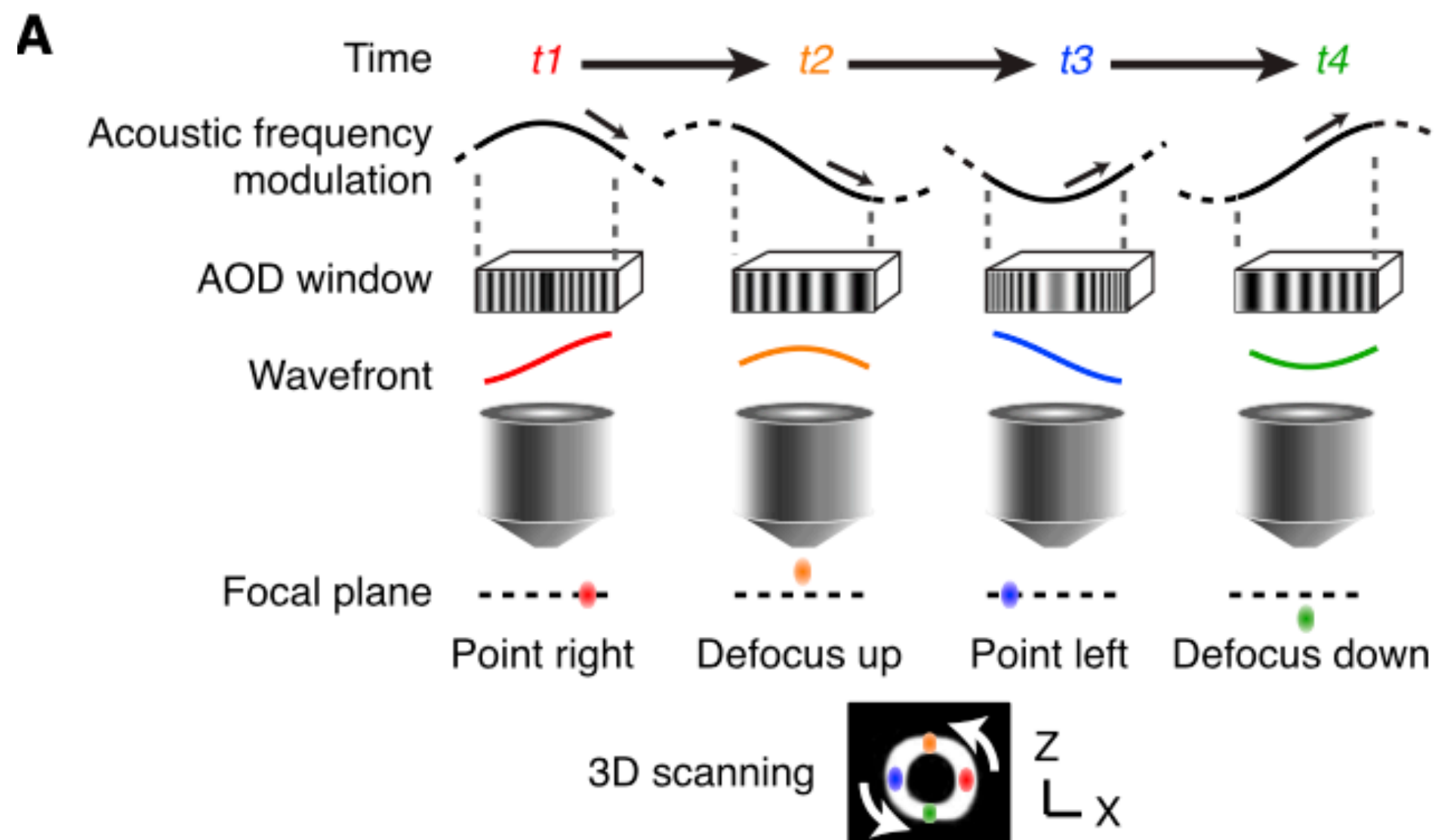


Griffiths et al., *Nat Methods* (2020)

RAMP for GEVI

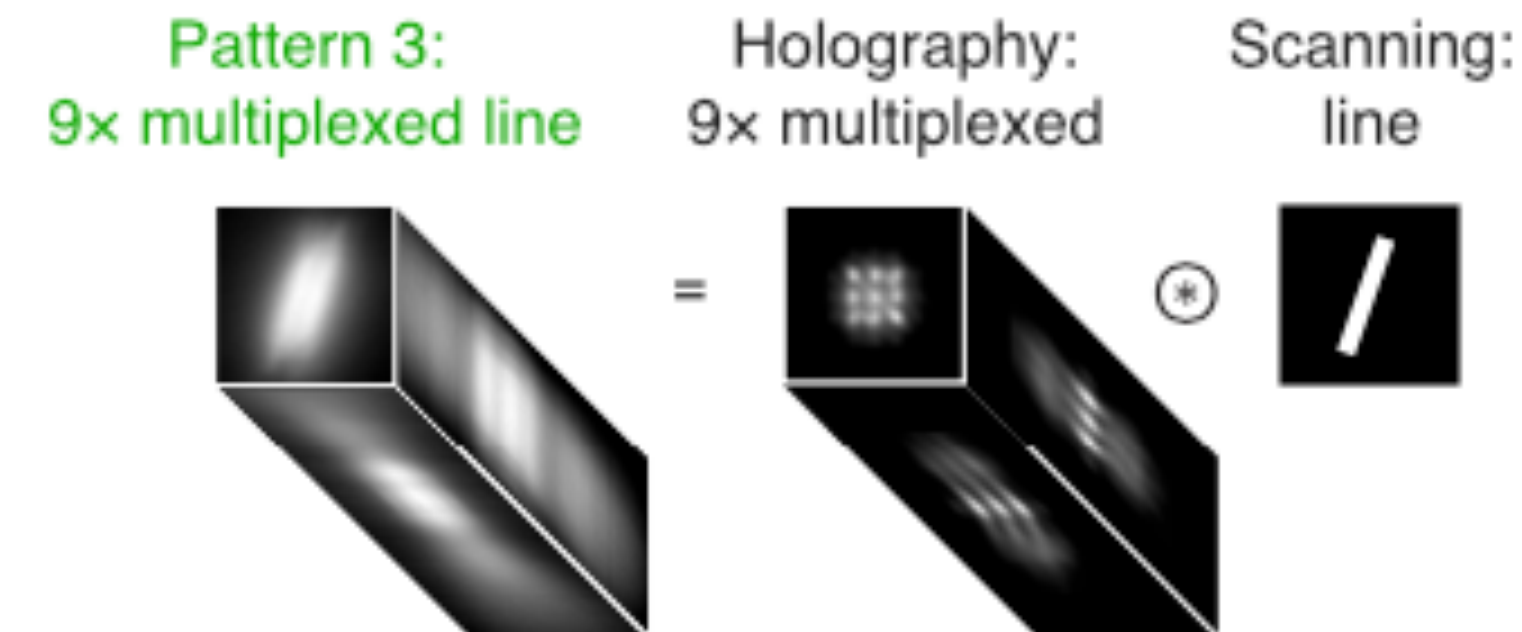
- Challenges:
 - membrane confined molecules → diffraction limited spot gives low signal
 - Slow diffusion limits replacement rates and causes photobleaching

ULoVe

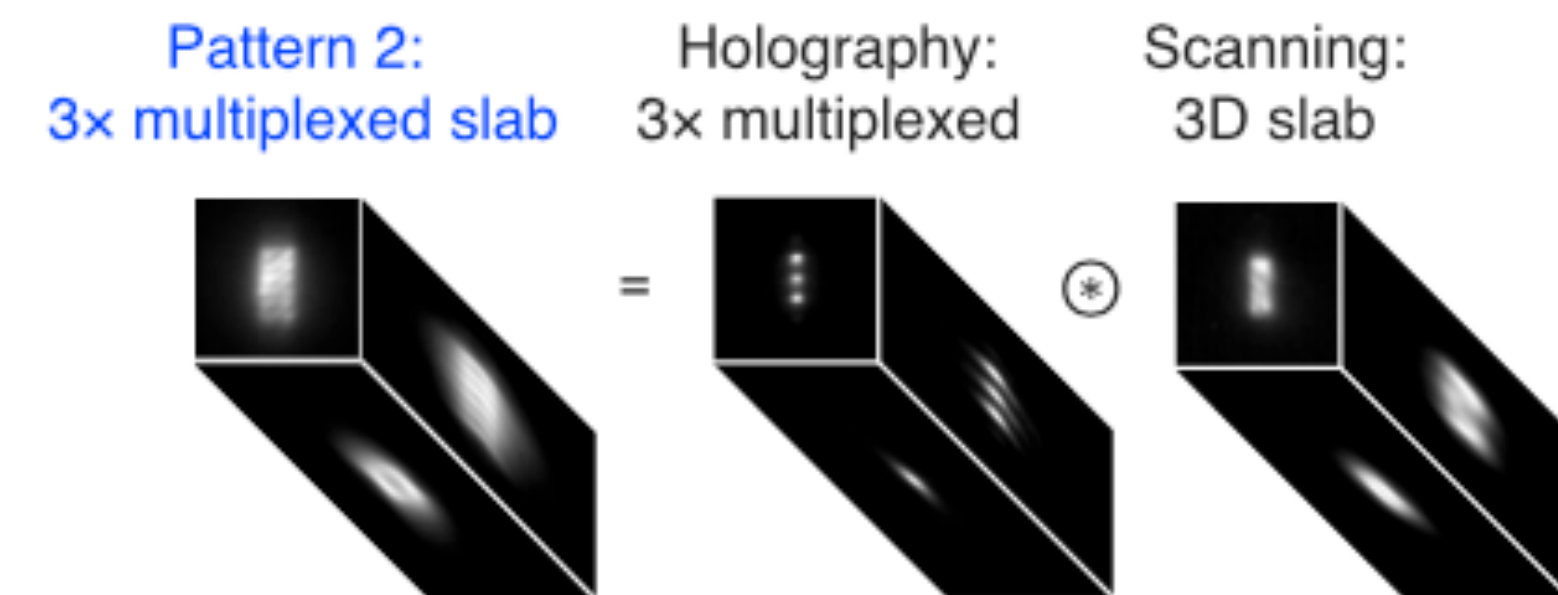


Ultrafast Two-Photon Imaging of a High-Gain Voltage Indicator in Awake Behaving Mice

Vincent Villette,^{1,13} Mariya Chavarha,^{2,3,13} Ivan K. Dimov,^{2,4} Jonathan Bradley,¹ Lagnajeet Pradhan,^{2,4} Benjamin Mathieu,¹ Stephen W. Evans,² Simon Chamberland,⁷ Dongqing Shi,^{2,5} Renzhi Yang,^{6,11} Benjamin B. Kim,³ Annick Ayon,¹ Abdelali Jalil,⁹ François St-Pierre,⁸ Mark J. Schnitzer,^{4,12} Guoqiang Bi,^{5,10} Katalin Toth,⁷ Jun Ding,⁶ Stéphane Dieudonné,^{1,*} and Michael Z. Lin^{2,3,14,*}

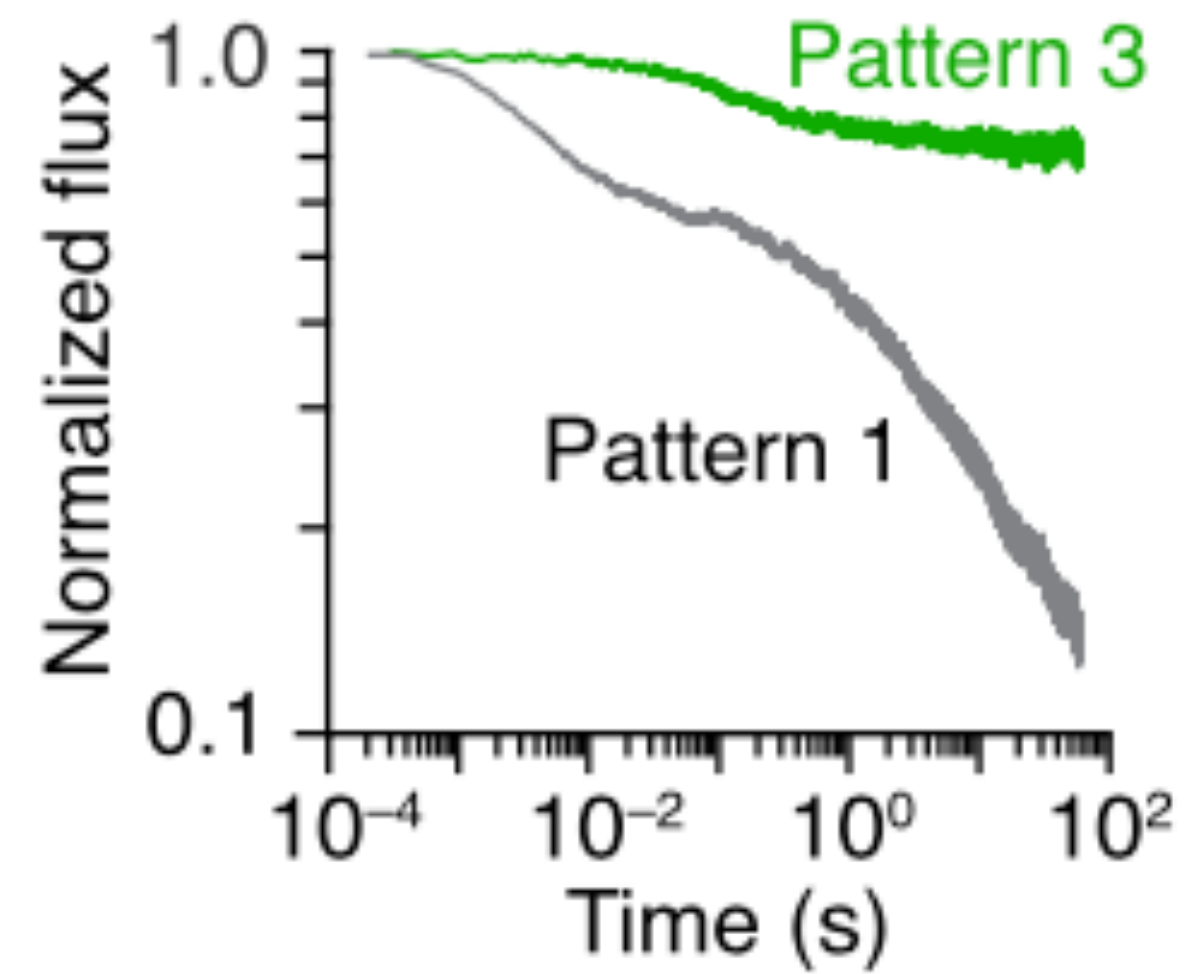


Akemann, W. Et al., Opt. Express 23, 28191–28205.

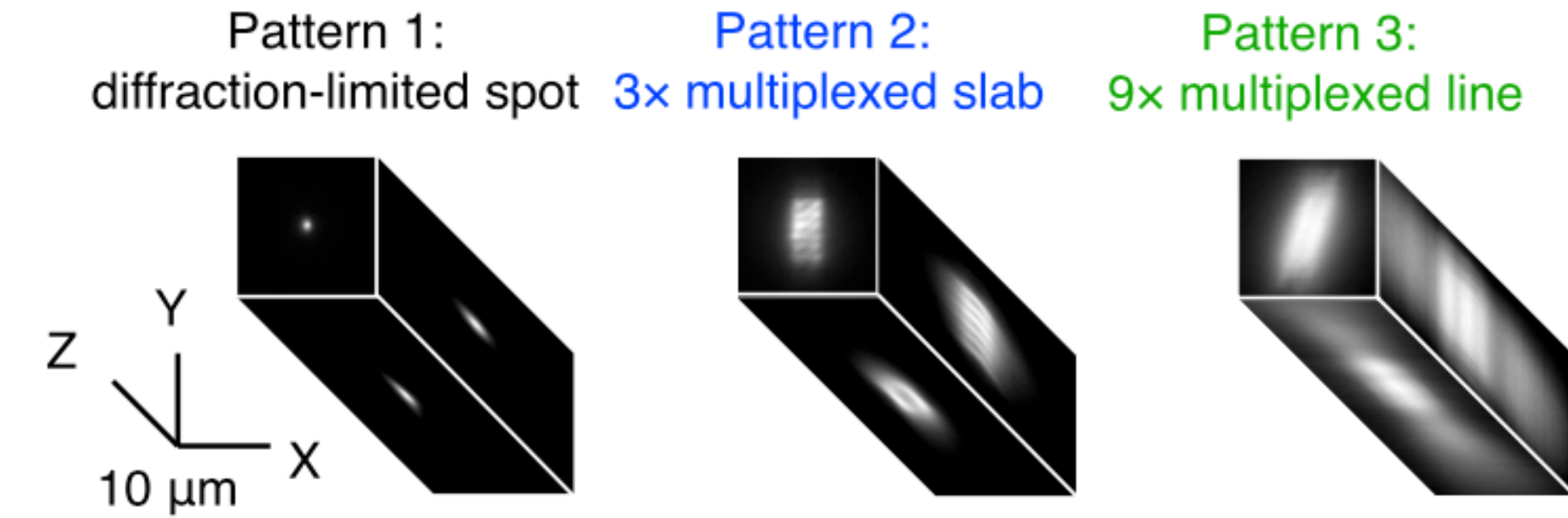
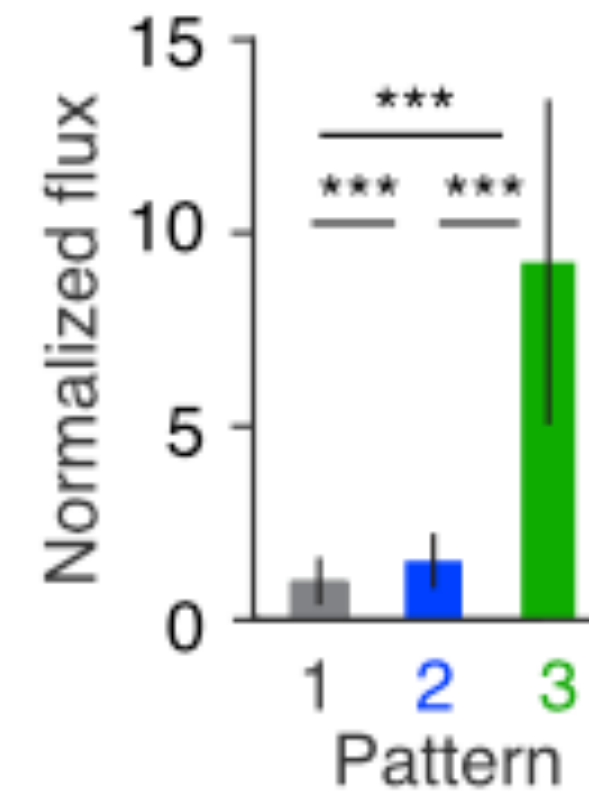


Villette et al., (2019)

Photobleaching:

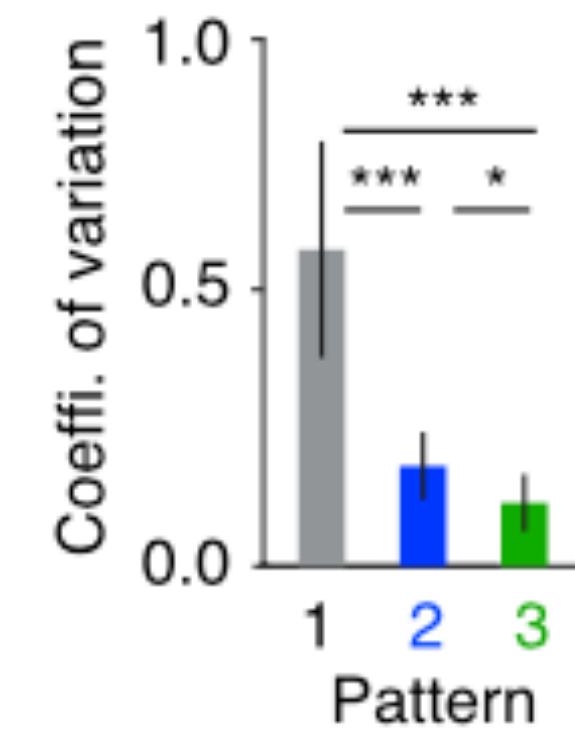
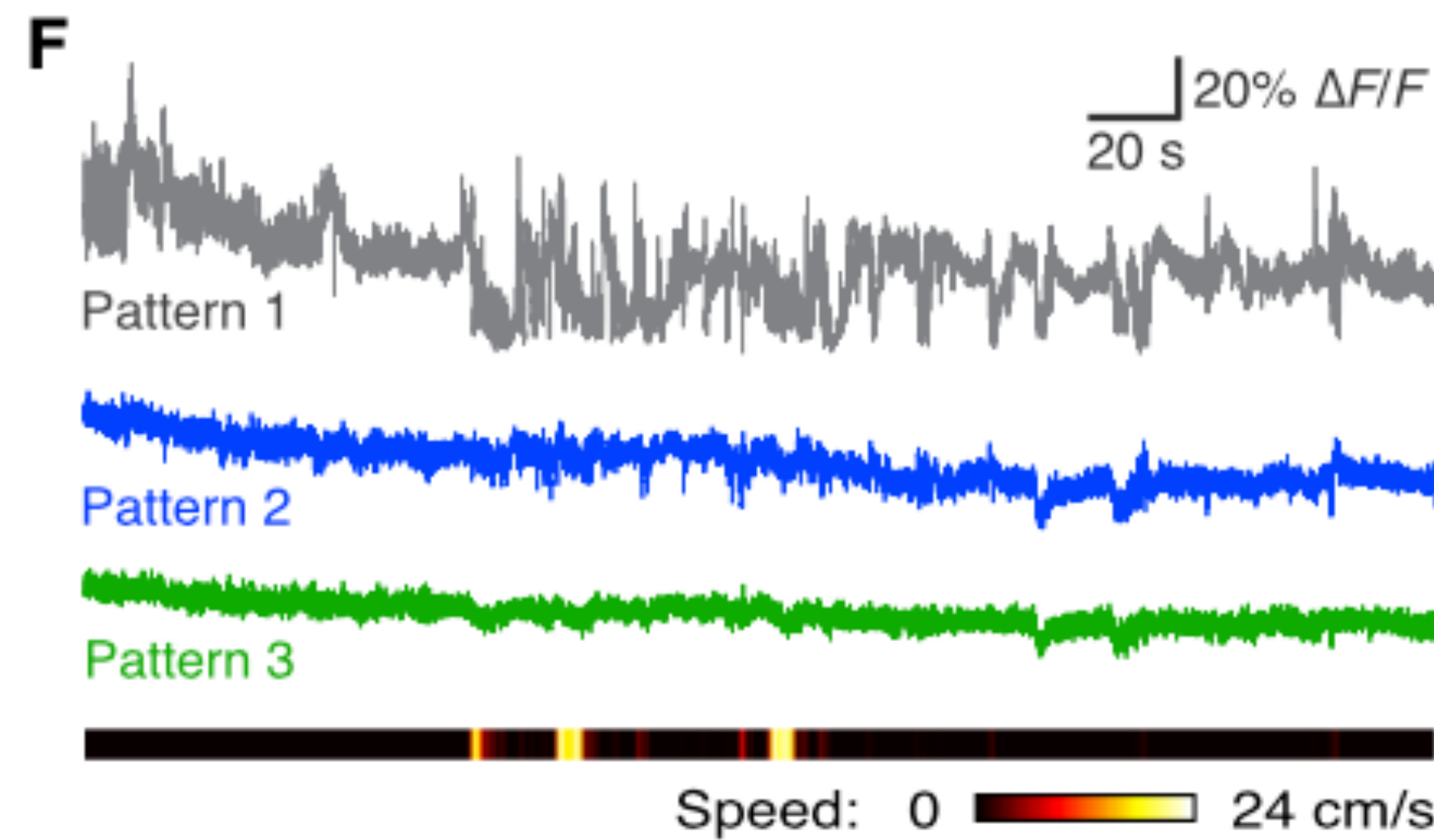
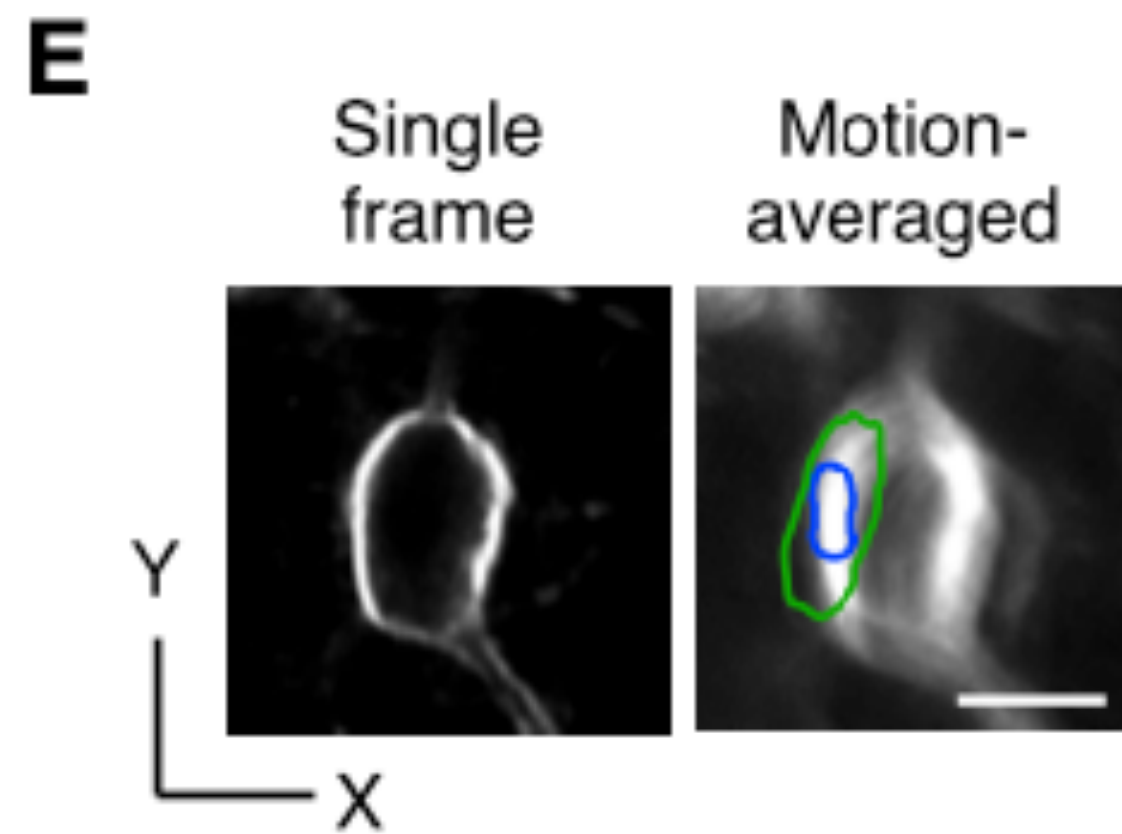


Signal:

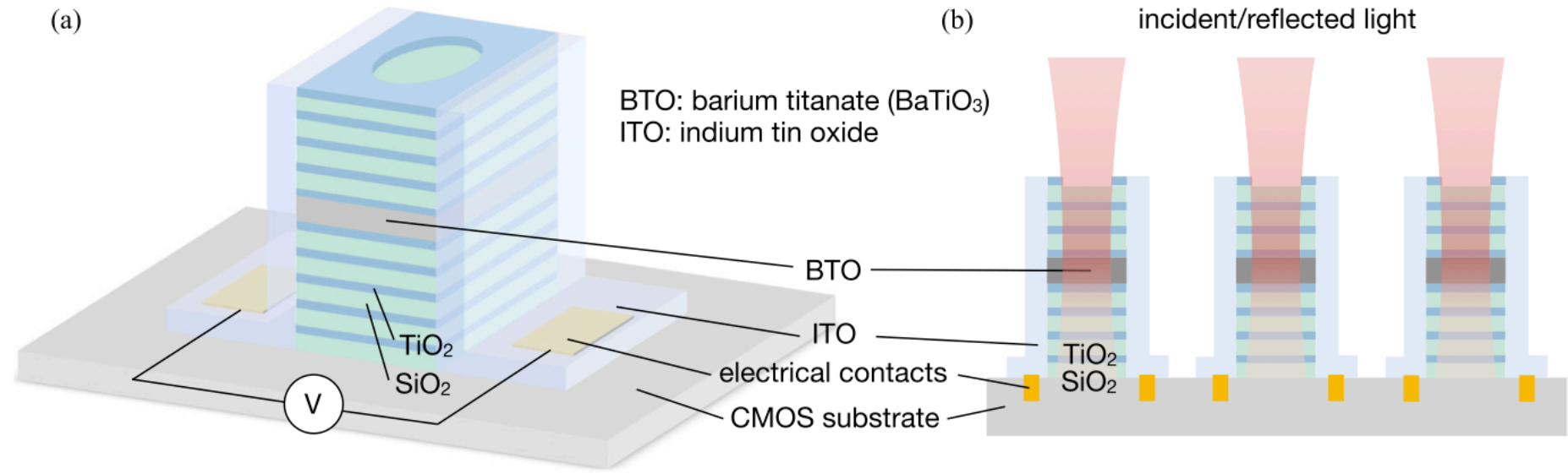


- + Increased signal
- + Reduced photobleaching
- + Movement correction
- 20 KHz per volume (cell)

Movement:



(a)



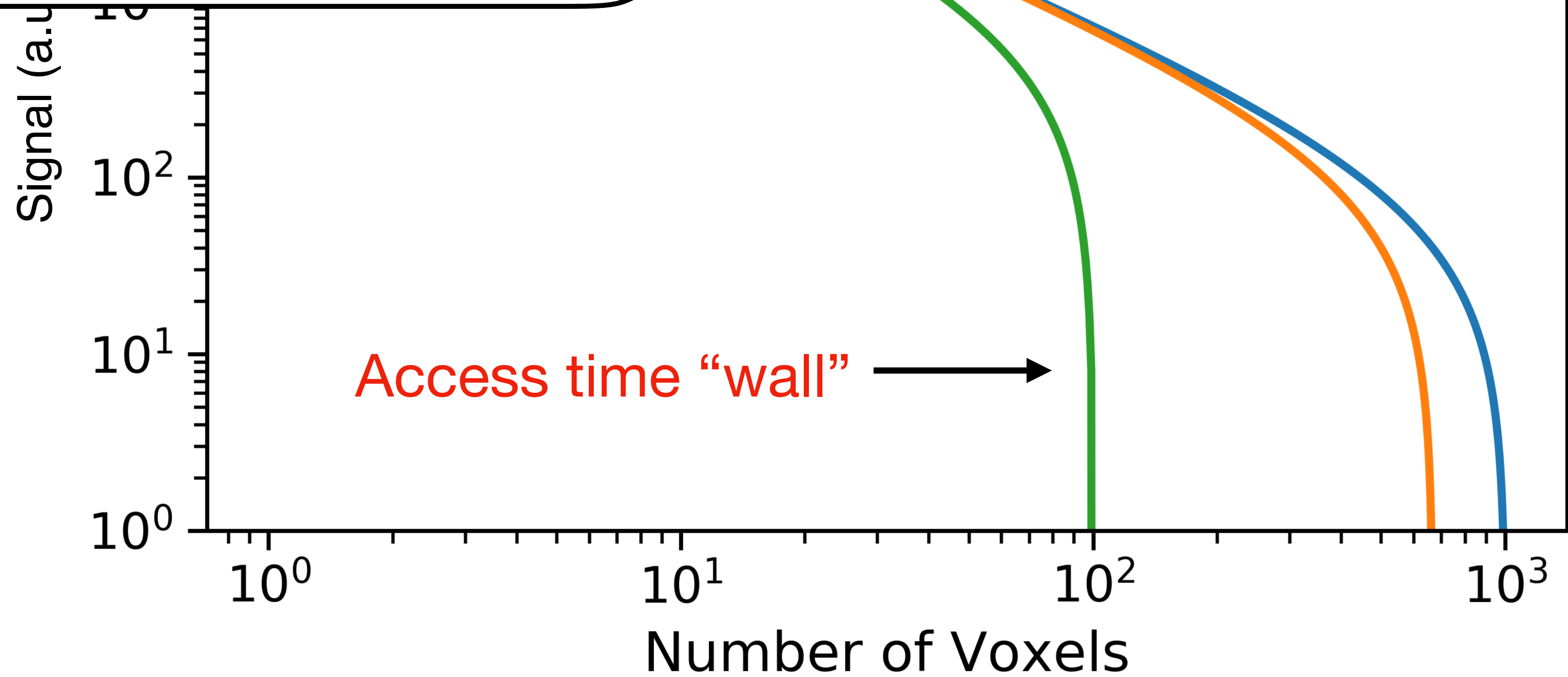
Design of high-speed phase-only spatial light modulators with two-dimensional tunable microcavity arrays

CHENG PENG,¹  RYAN HAMERLY,¹ MOHAMMAD SOLTANI,² AND DIRK R. ENGLUND^{1,*}

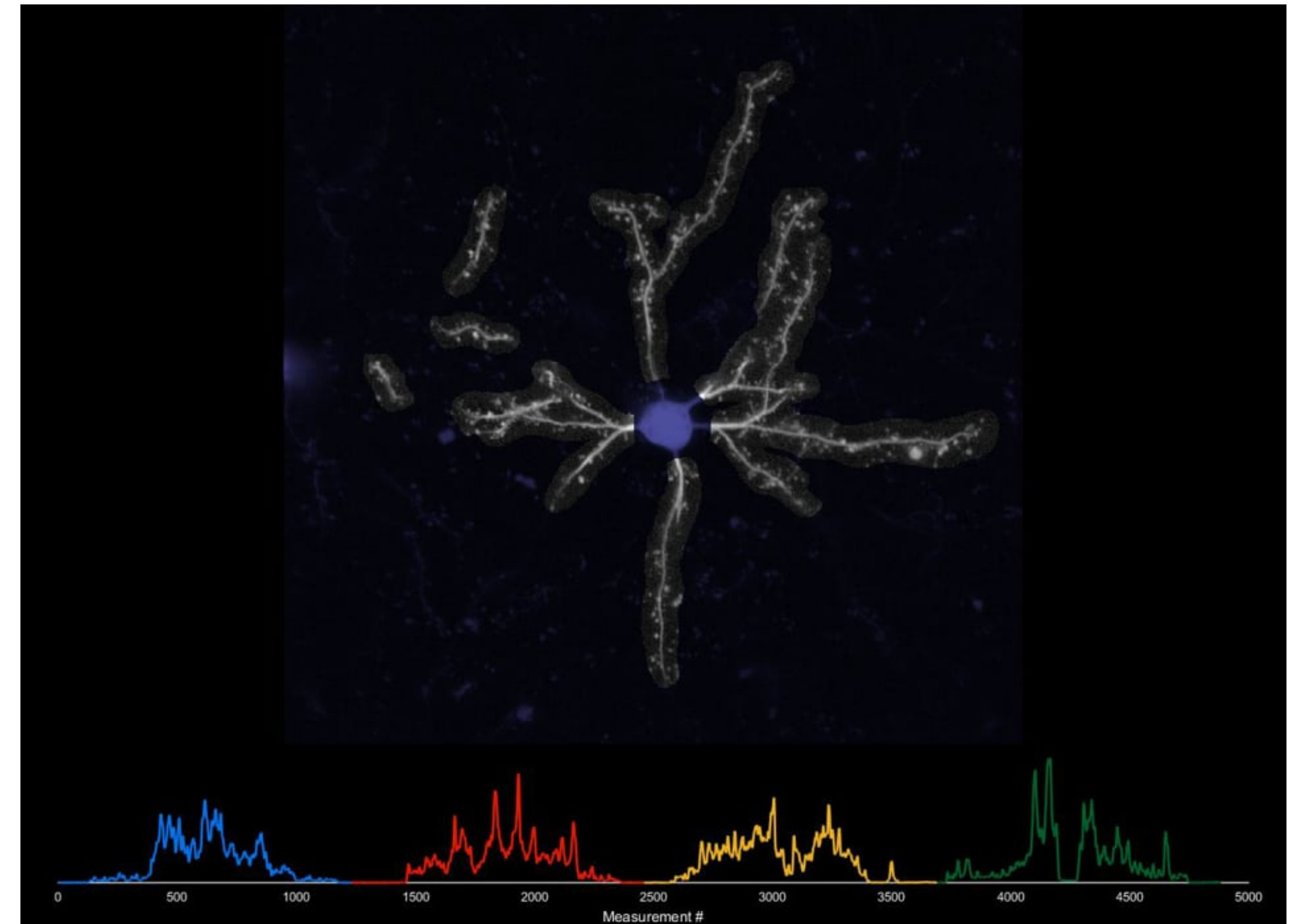
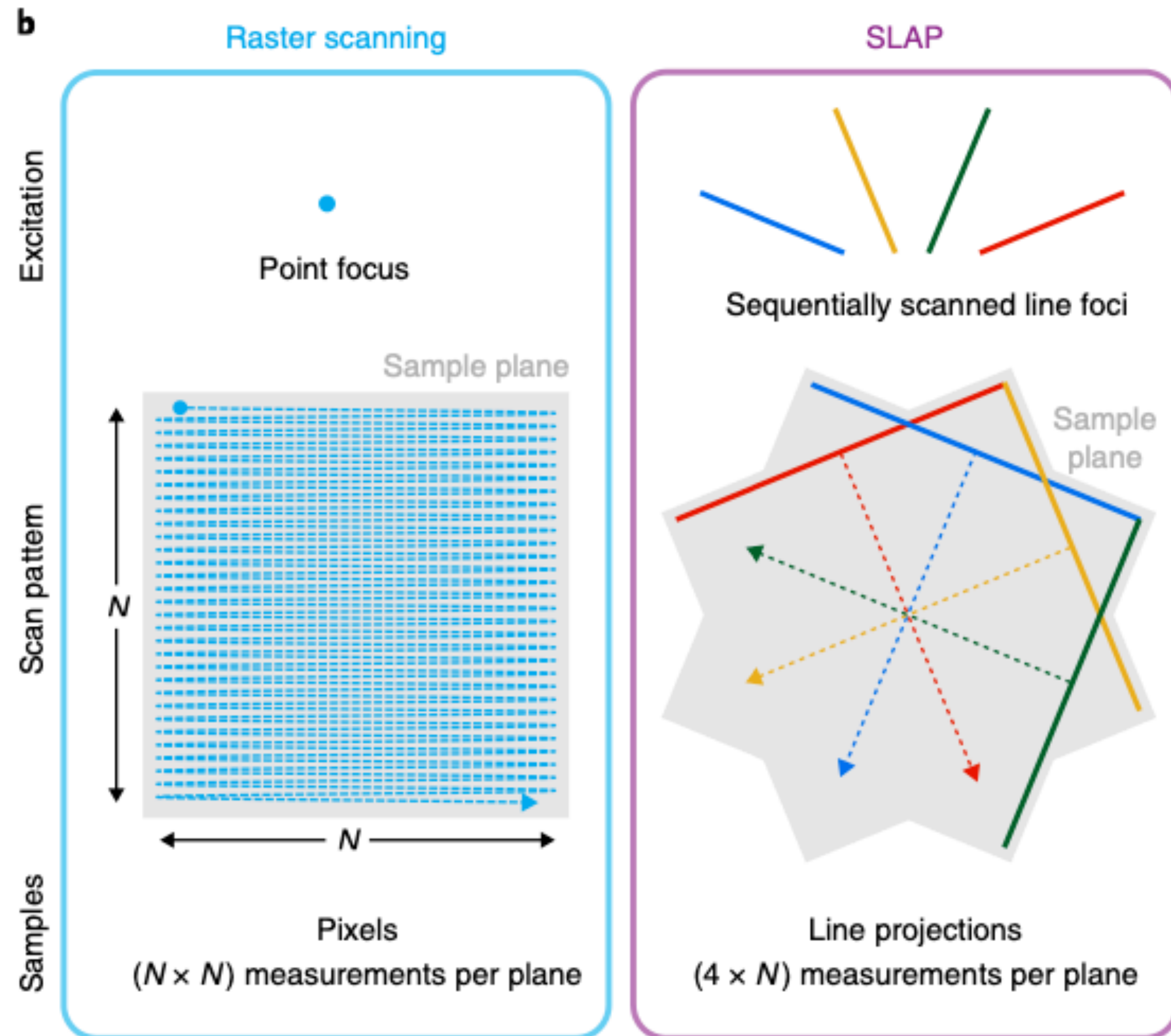
(b)

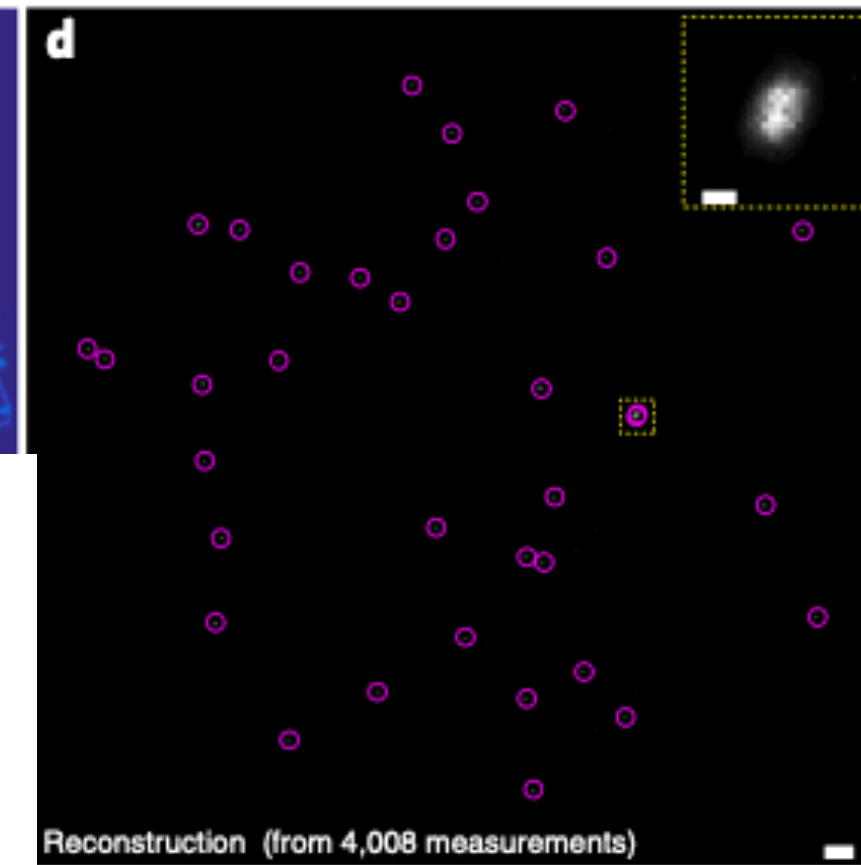
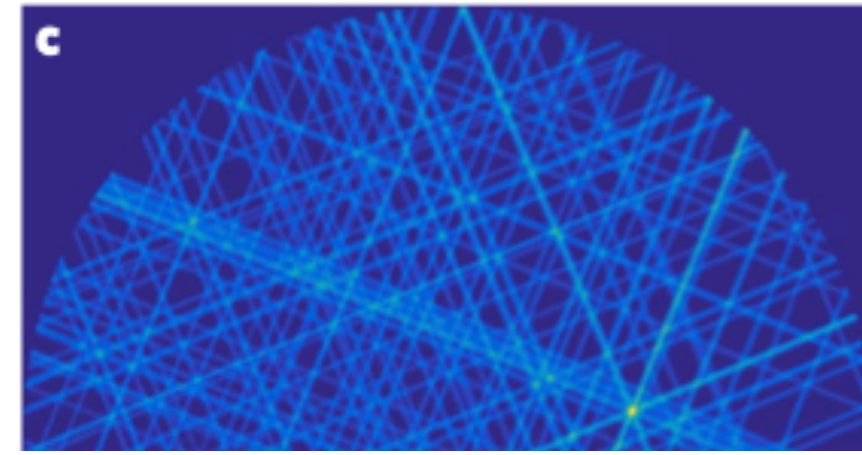
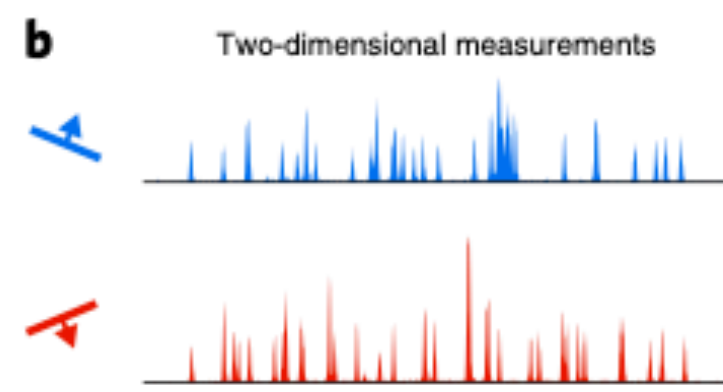
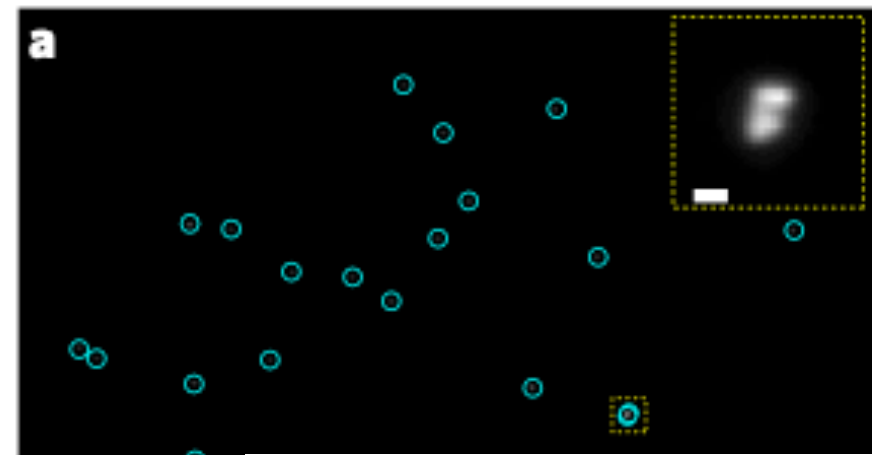
incident/reflected light

1.0e-06 μ s
1.5e-06 μ s
1.0e-05 μ s



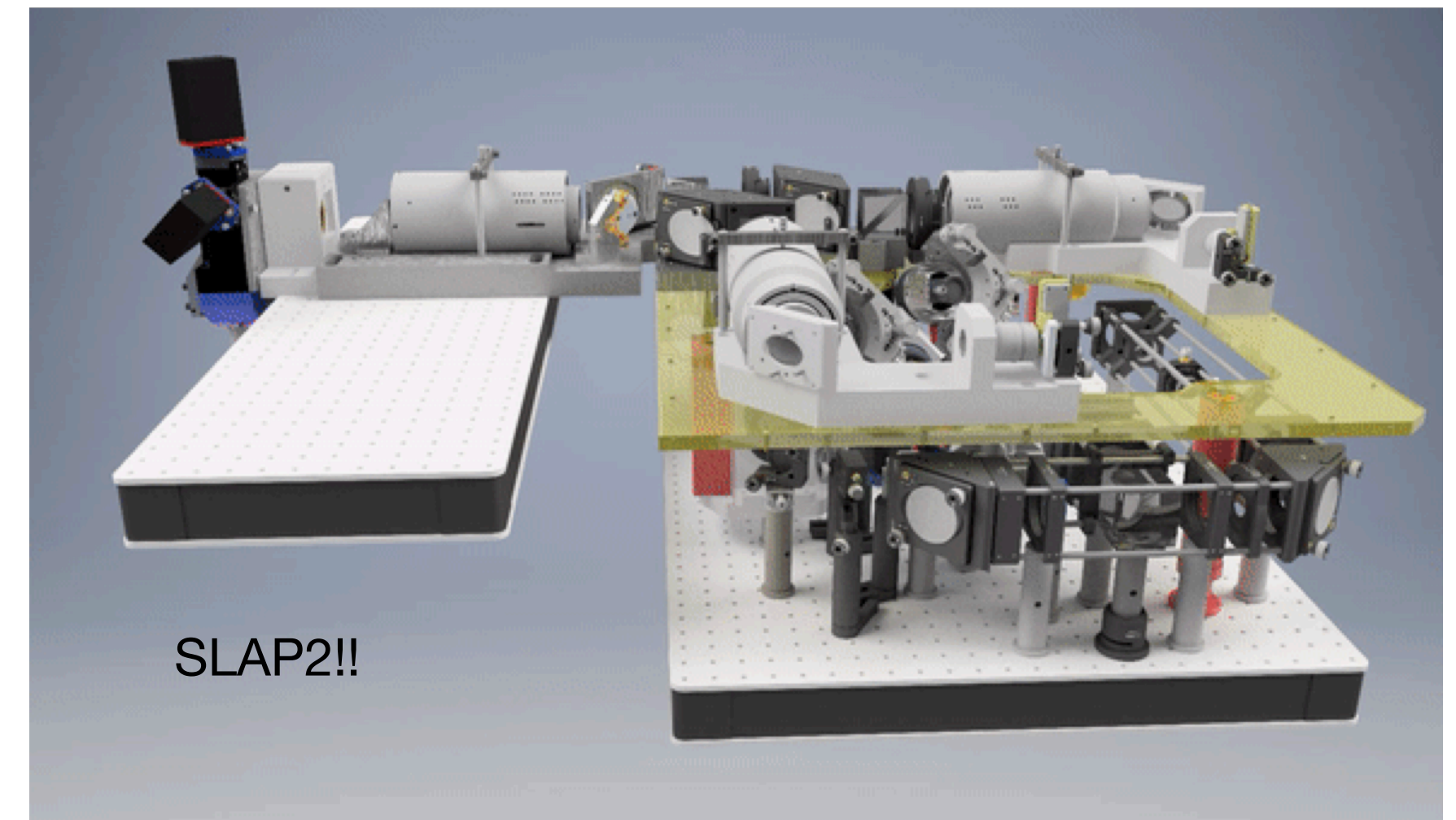
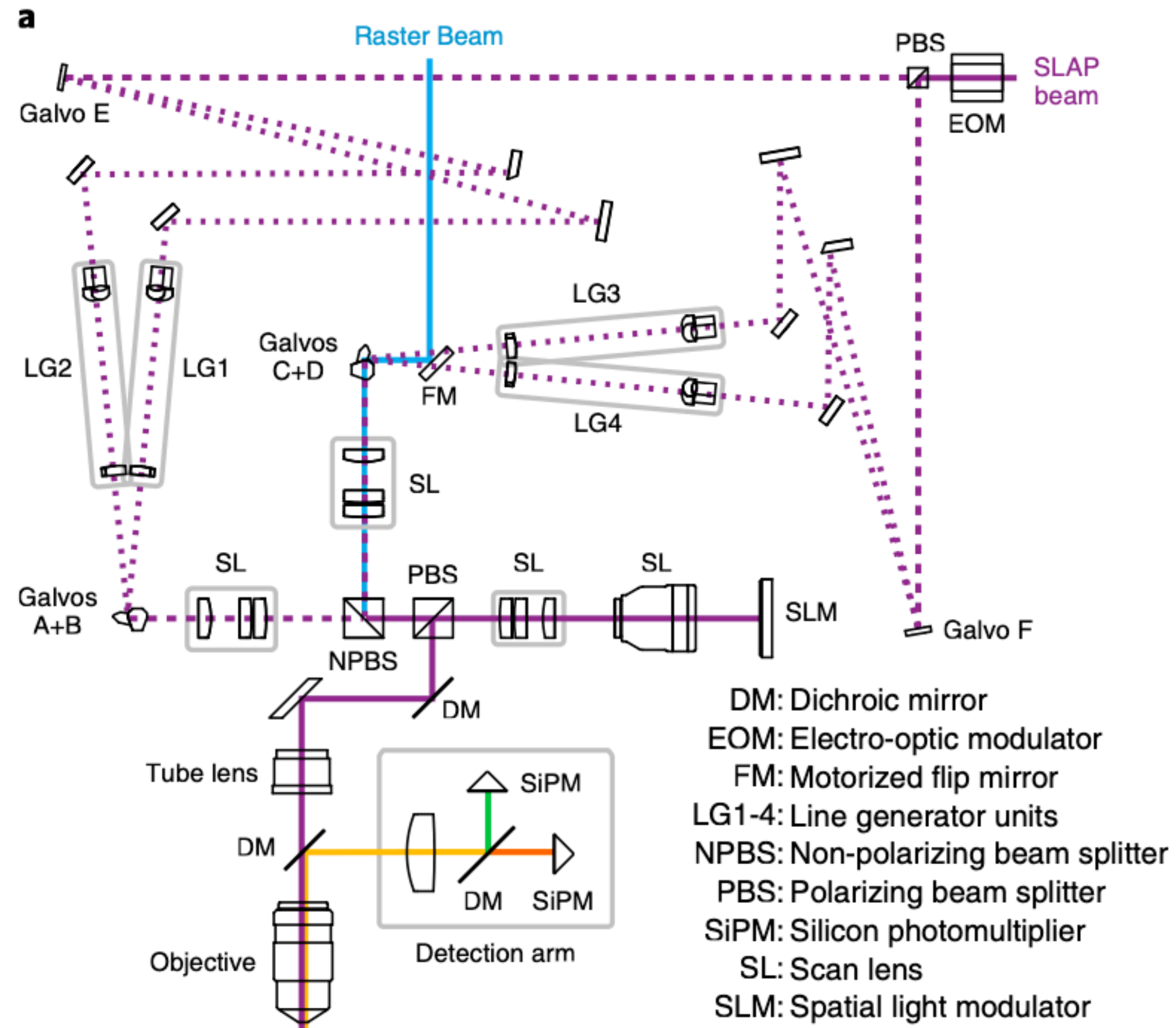
Computational Imaging (SLAP)





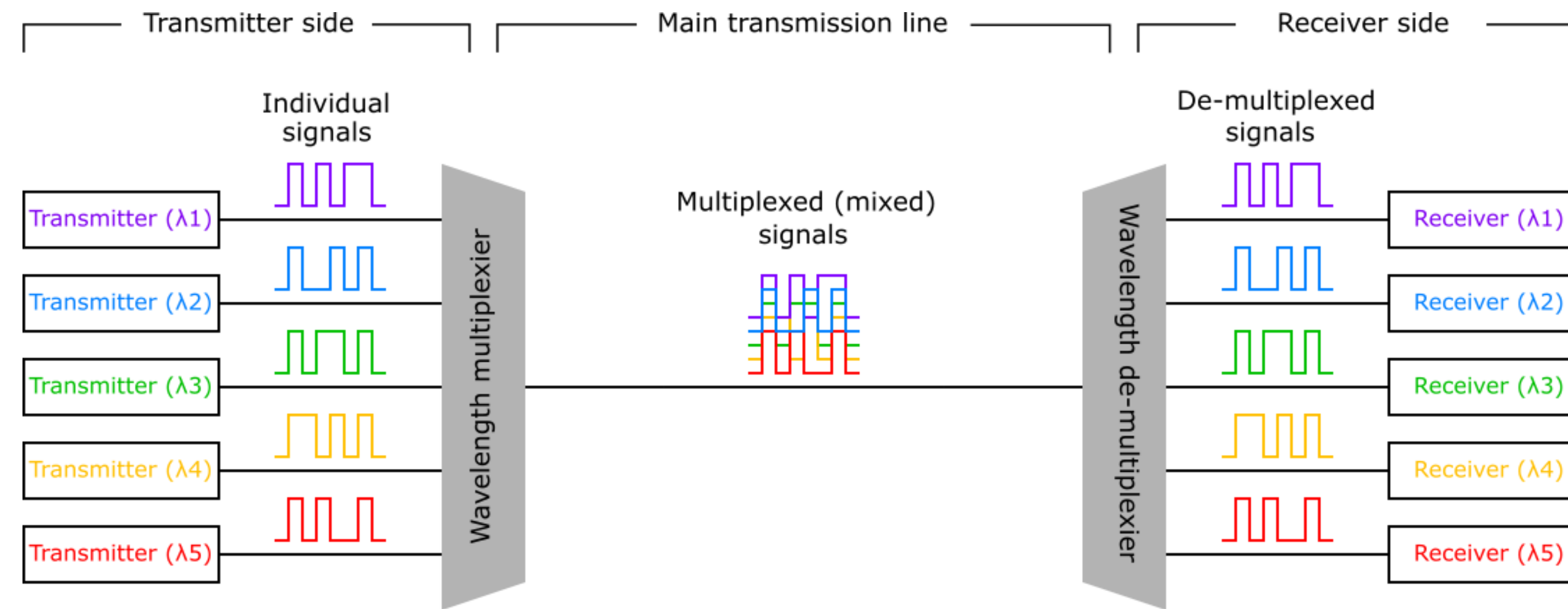
- + Fast
- + Dense or sparse
- + Robust to movement
- + Accurate reconstruction
- Power hungry!
- Complex
- Limited FOV ($\sim 250 \mu\text{m}$)
- Requires a prior

1m res.)



<https://www.janelia.org/lab/podgorski-lab/slap2>

Multiplexing



**Generate many beams
(x, y, z)**

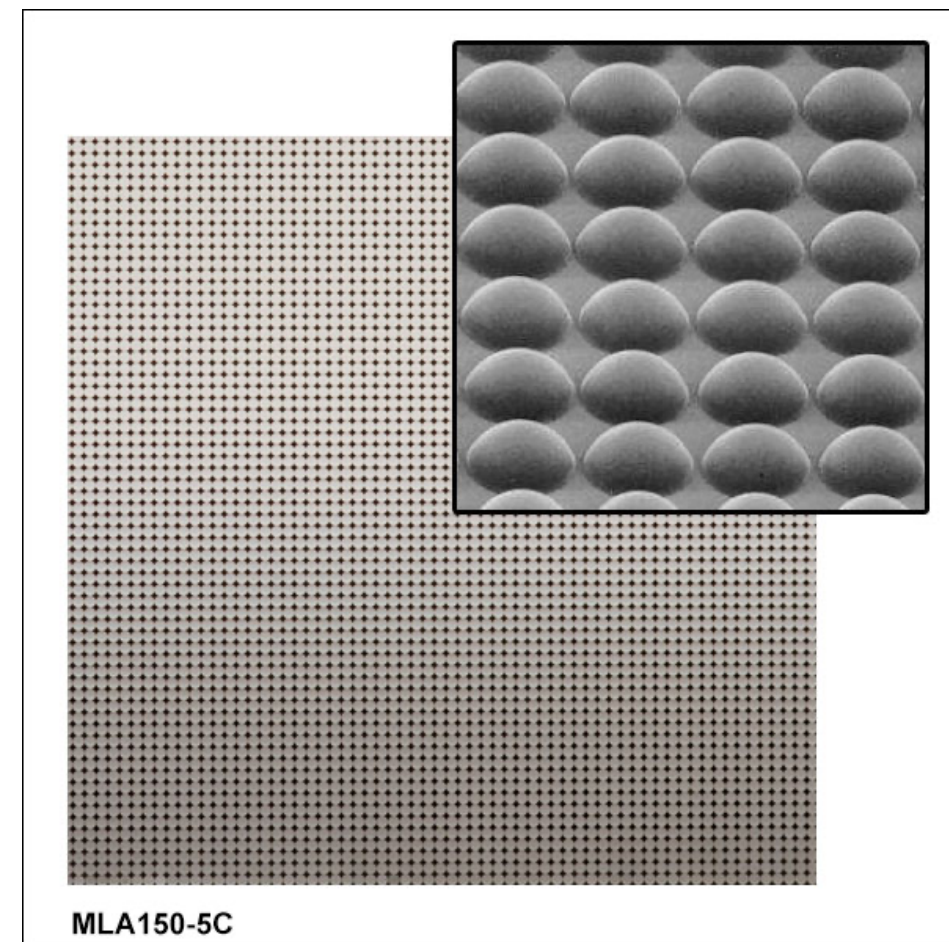


**Scan sample
in parallel**

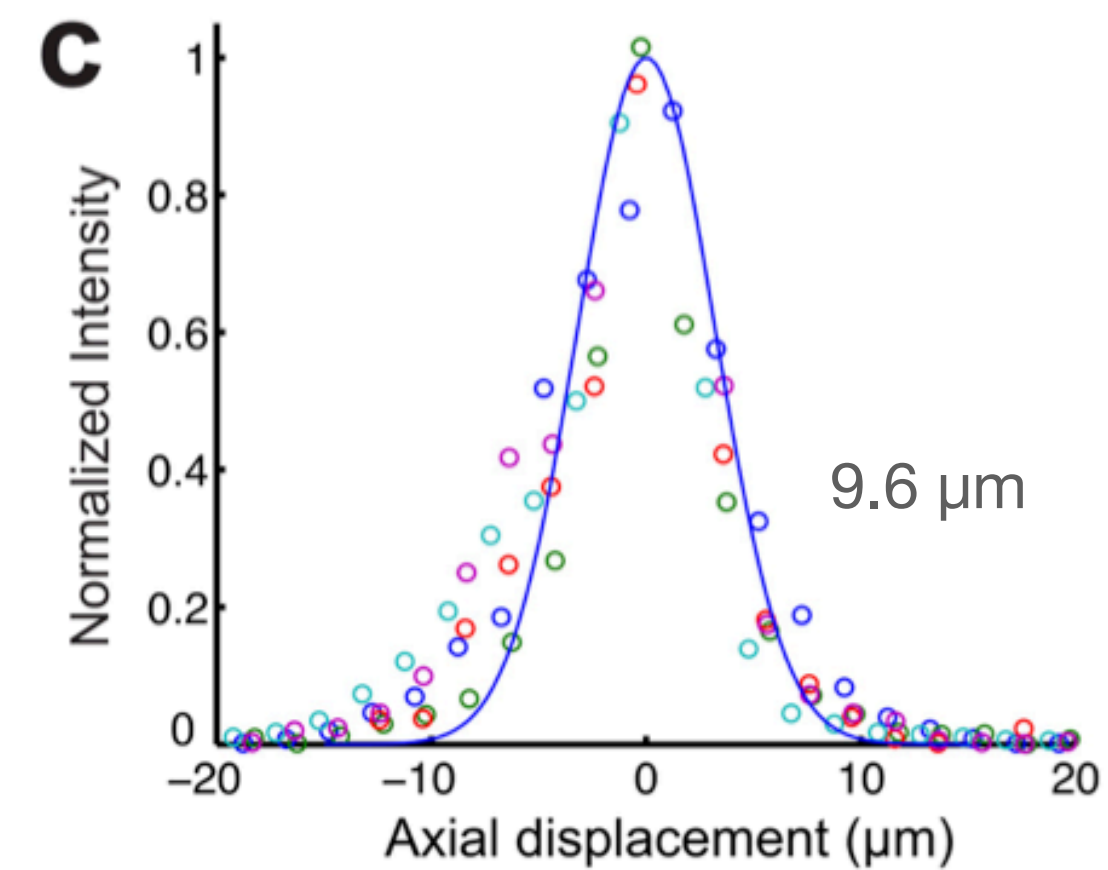
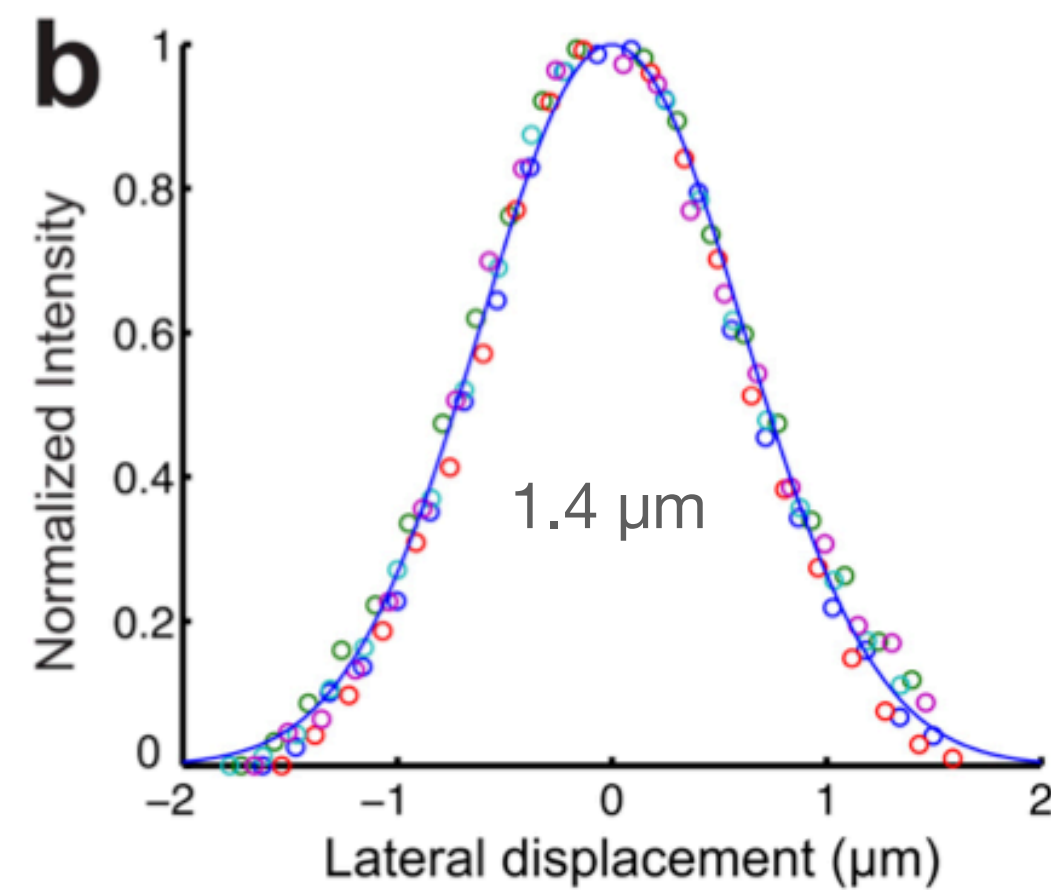
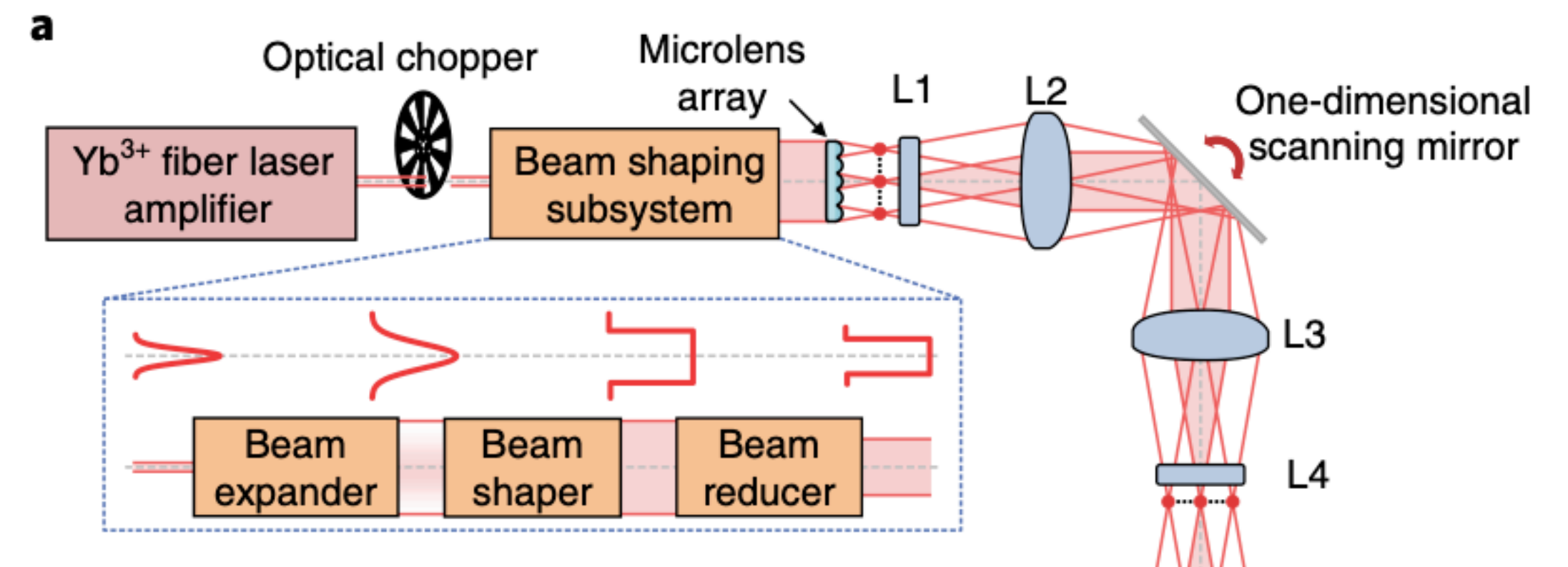
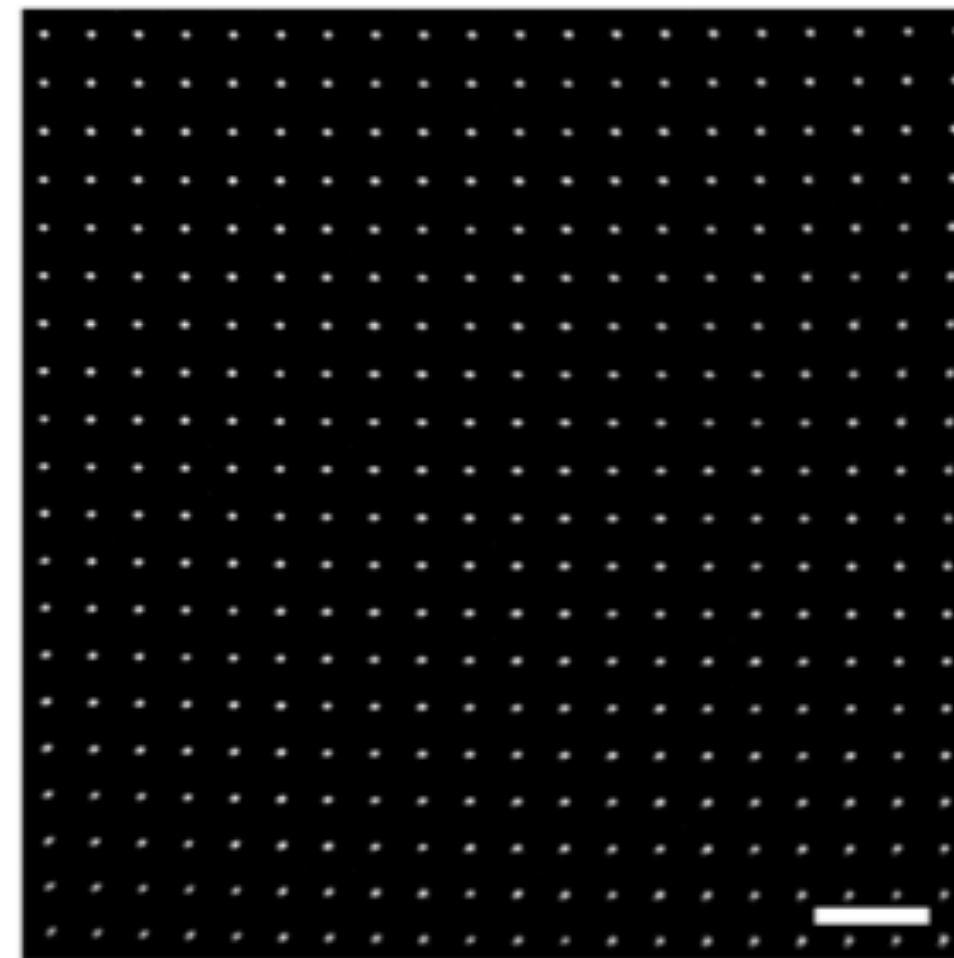


**Separate and detect fluorescent signals
(time, space, freq)**

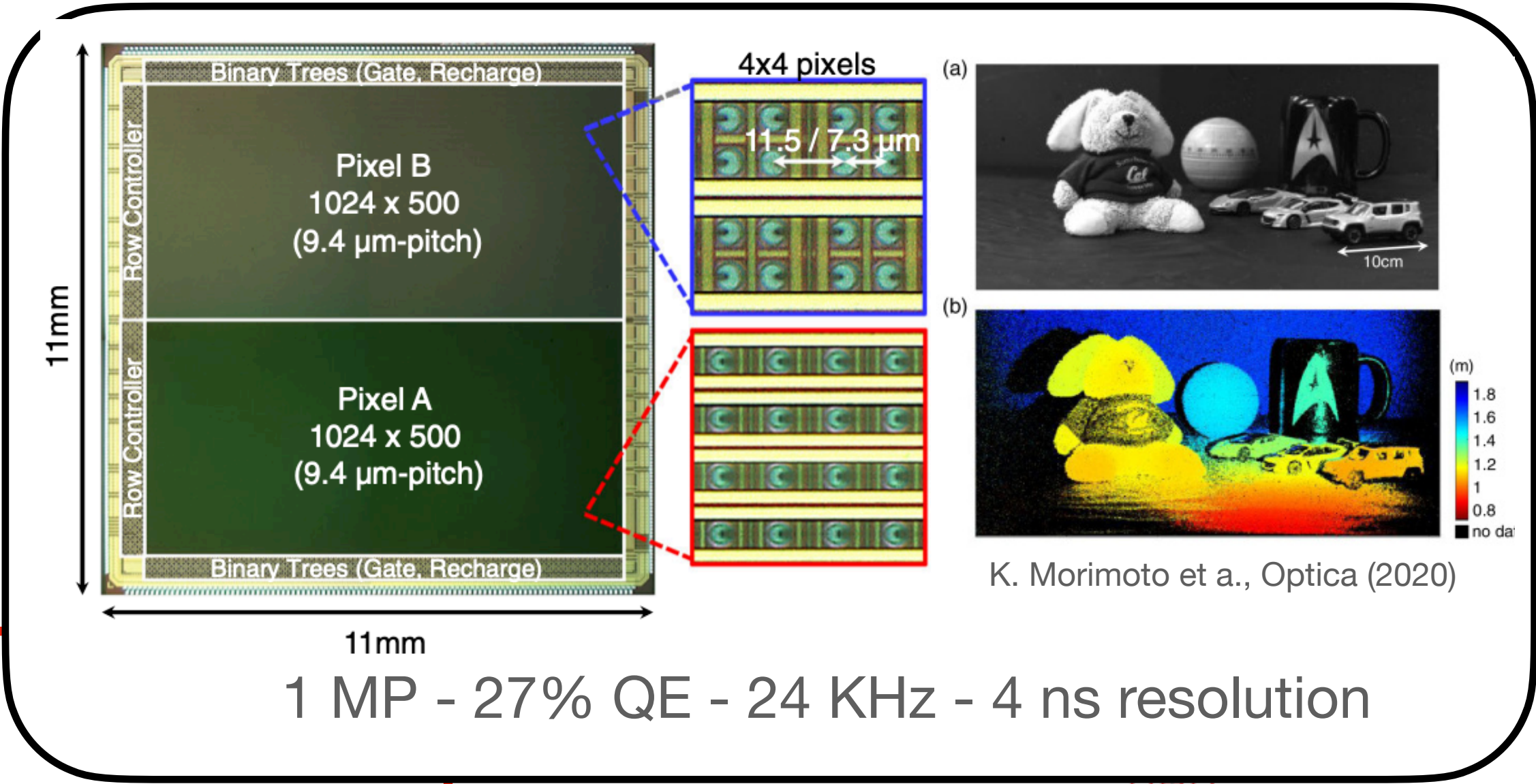
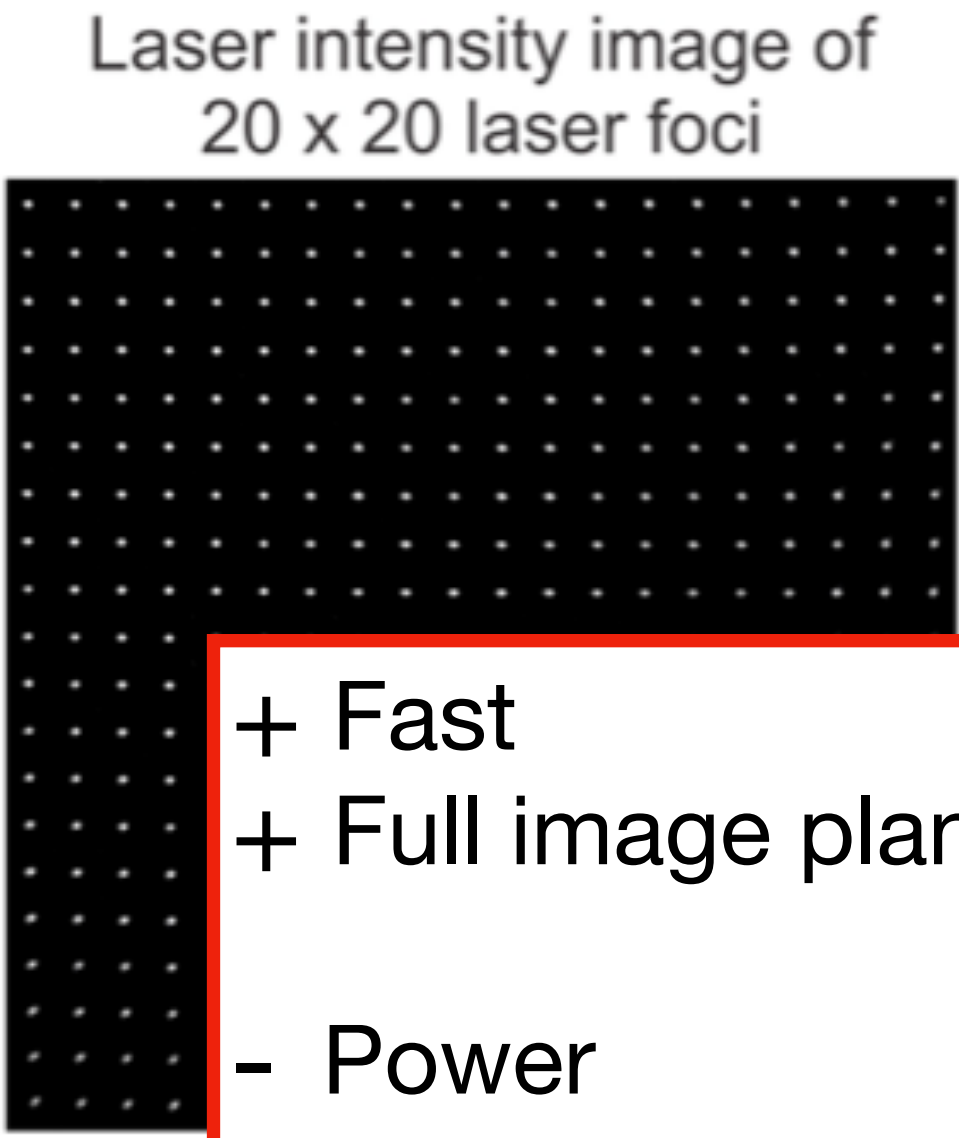
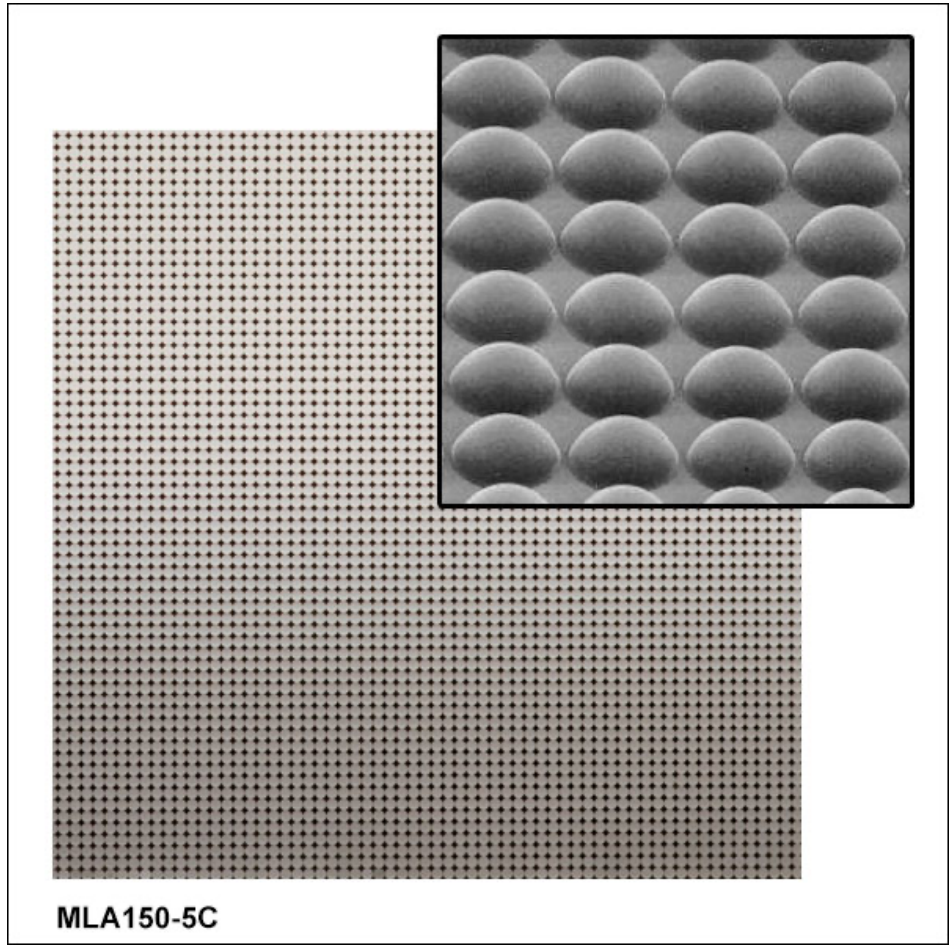
Spatial MUX



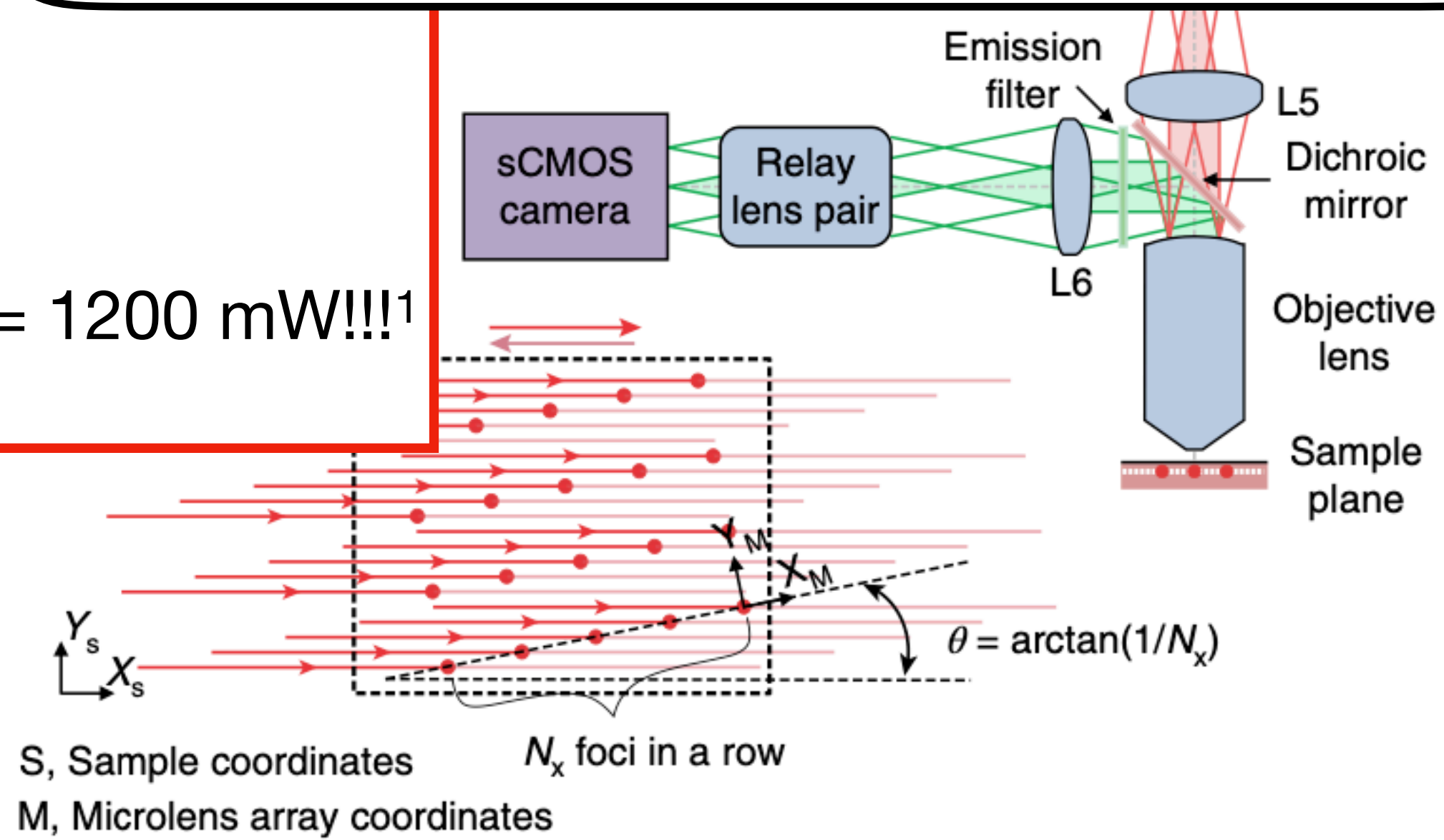
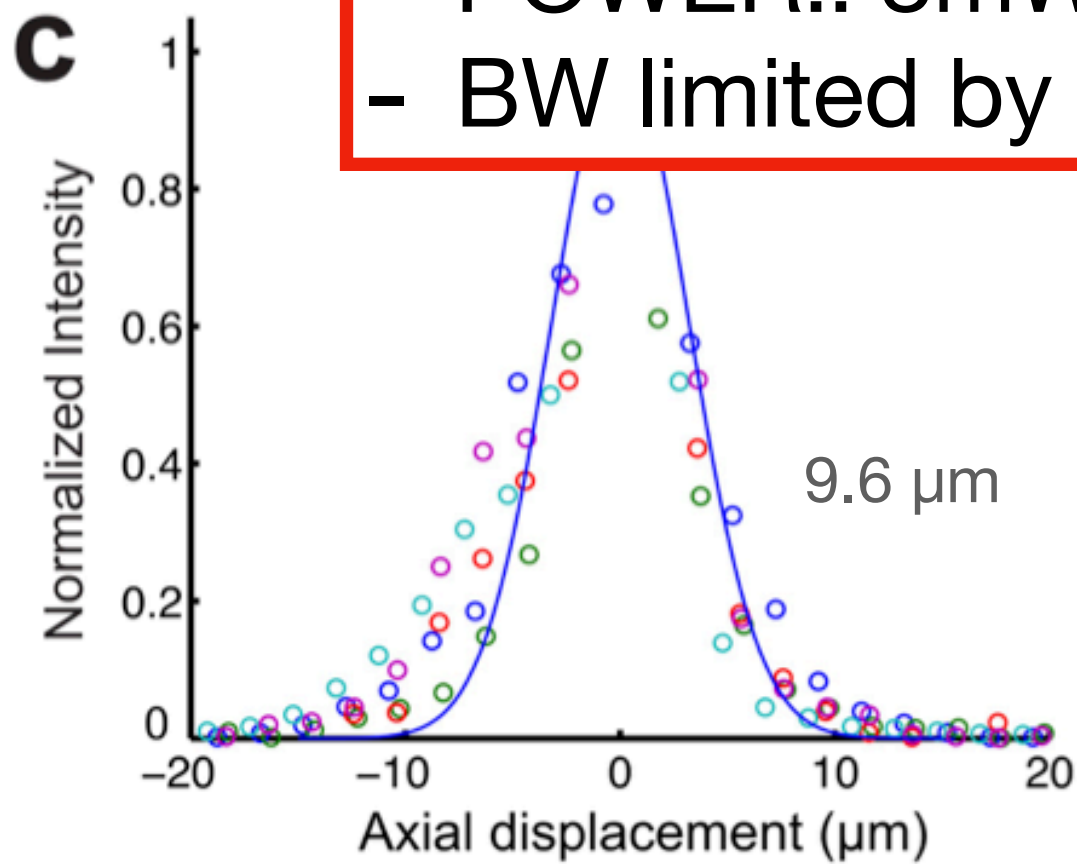
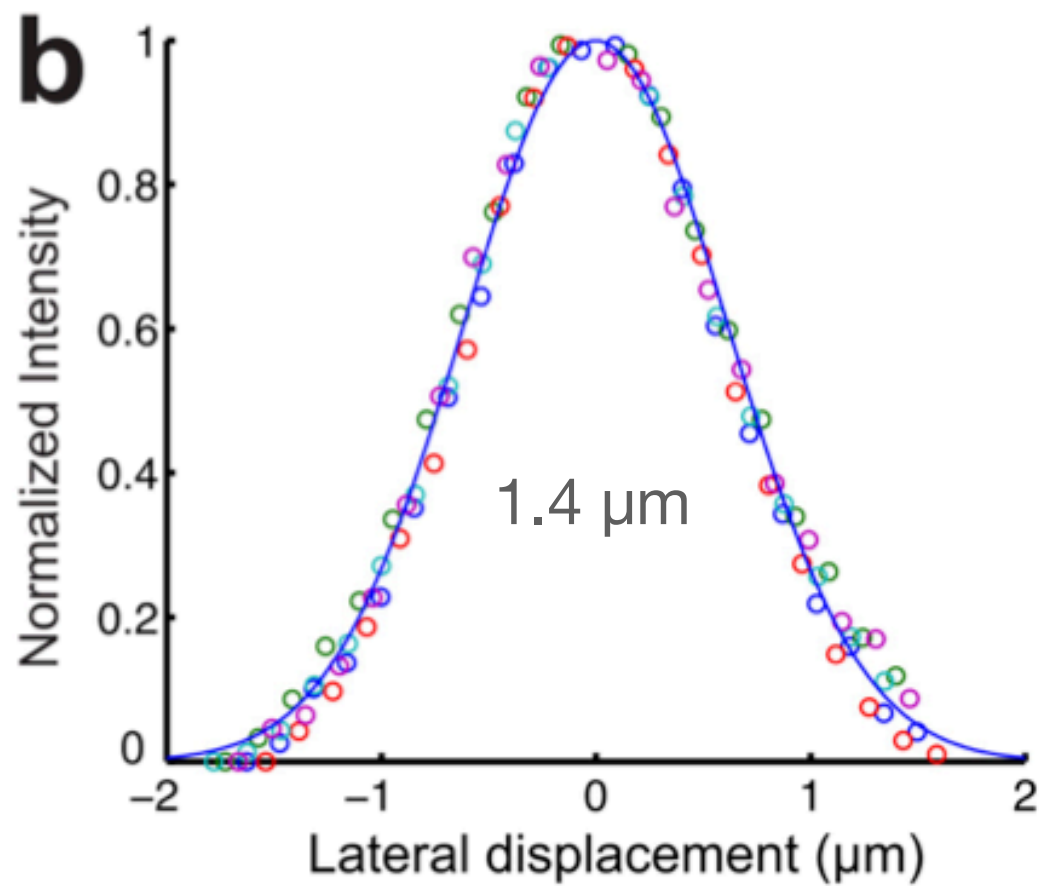
Laser intensity image of
20 x 20 laser foci



Spatial MUX



- + Fast
- + Full image plane
- Power
- Power!
- POWER!! 3mW x 400 = 1200 mW!!!!¹
- BW limited by camera

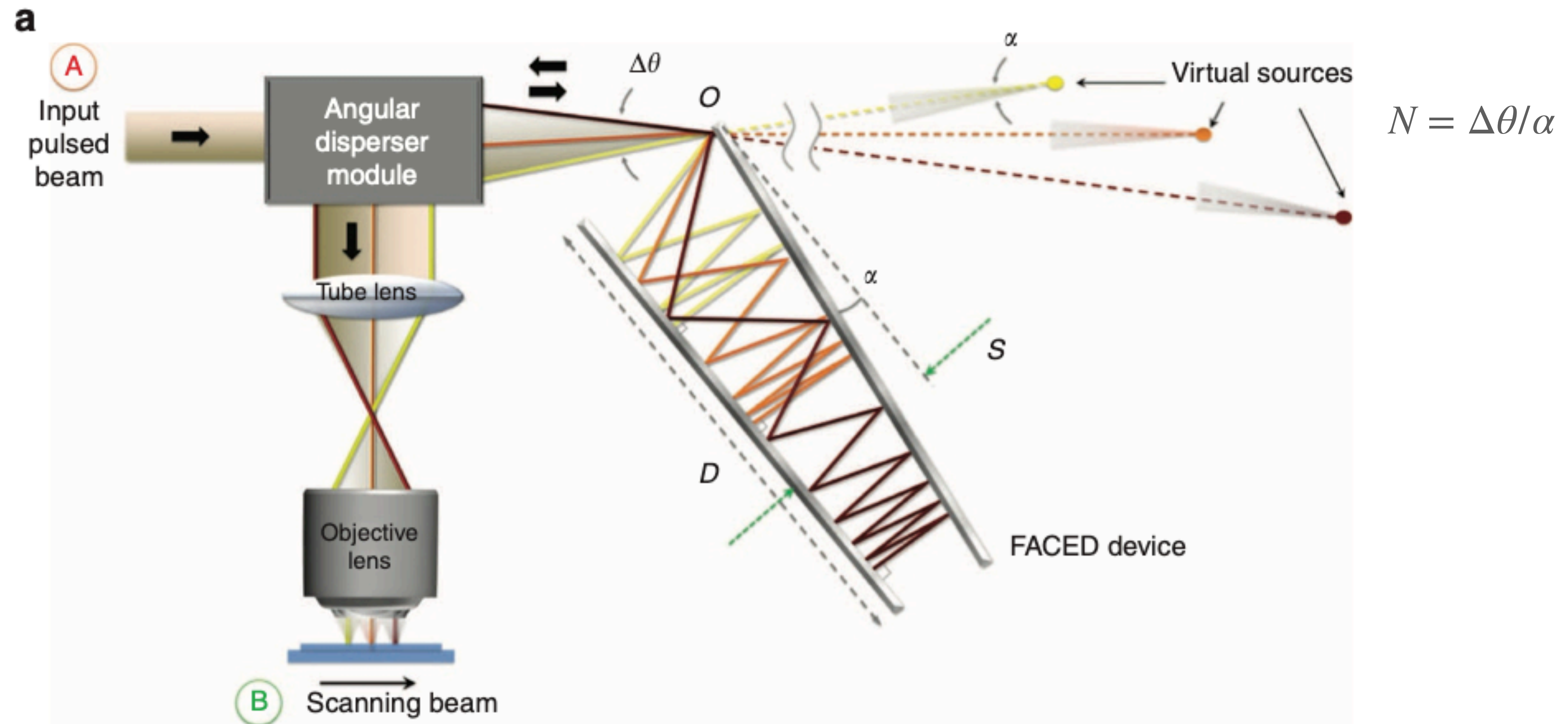


¹Podgorski and Ranganathan (2016)
[Max 250 mW]

Spatiotemporal MUX in X

FACED Microscopy

Line scanning at the rep-rate of the laser!! ~ MHz



J. Wu ... M. Z. Lin, K. K. Tsia, and N. Ji, *Nat Meth* (2020).

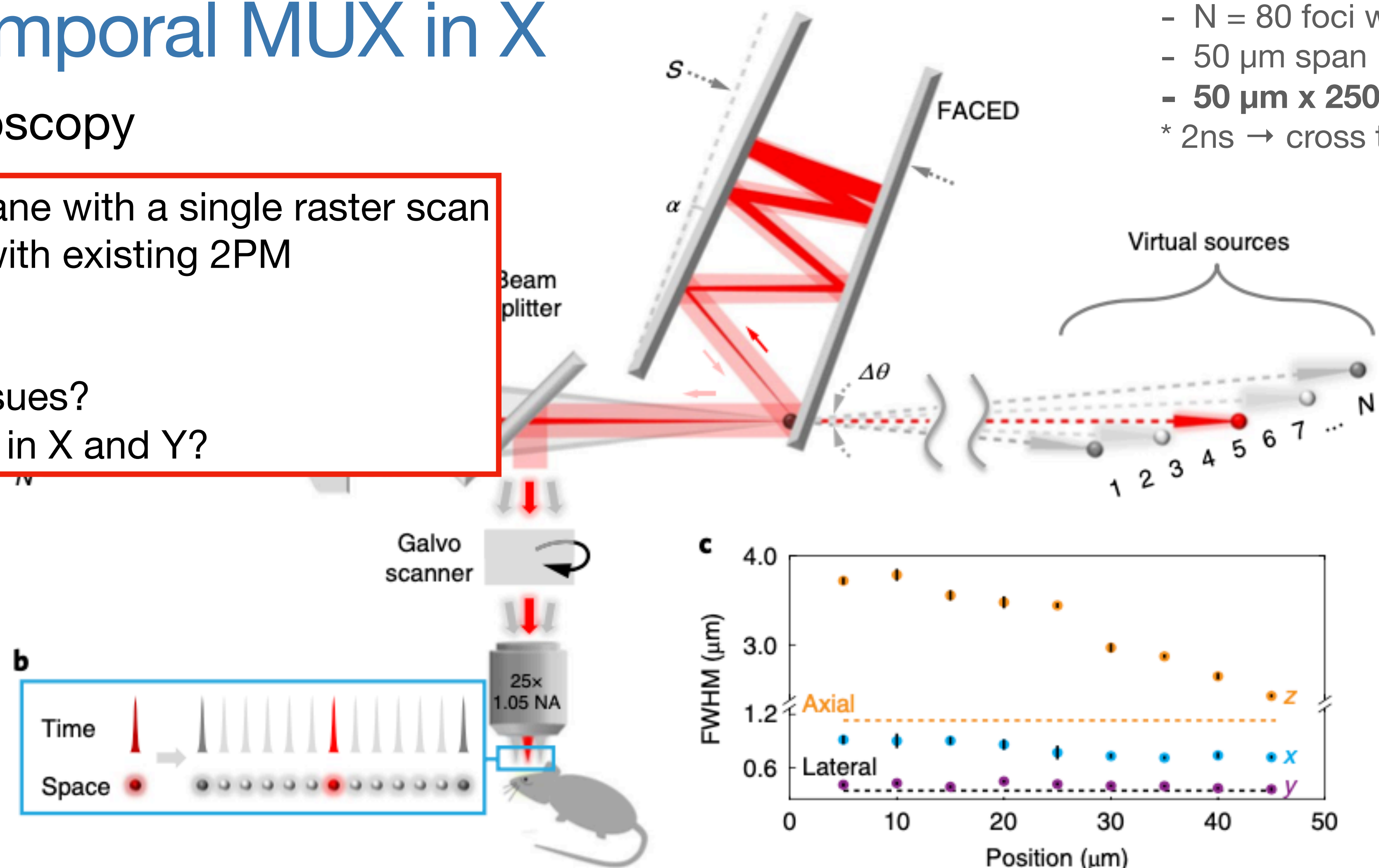
J.-L. Wu ... K. K. Tsia, *Light Sci Appl* 6, e16196 (2017).

Spatiotemporal MUX in X

FACED Microscopy

+ Full image plane with a single raster scan
+ Compatible with existing 2PM

- Limited FOV
 - Dispersion issues?
- two together in X and Y?



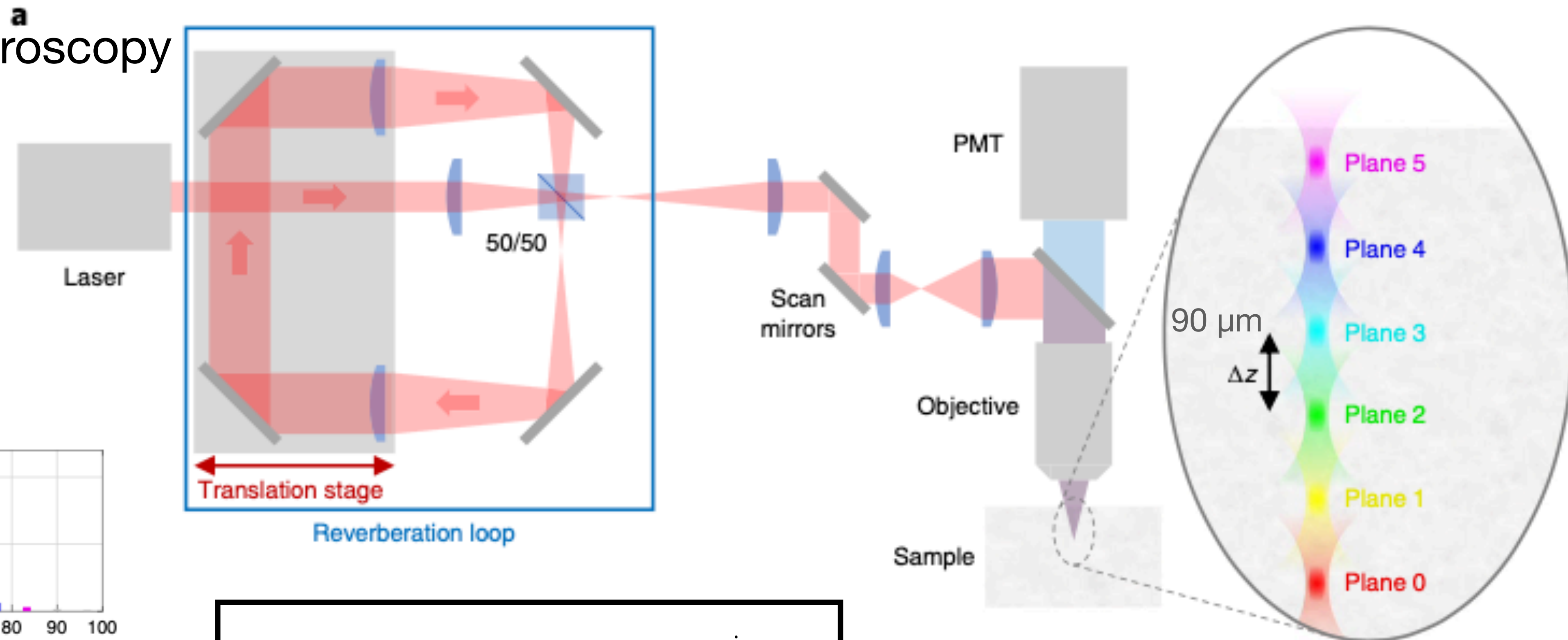
- 1 MHz laser
- $N = 80$ foci with 2 ns delay*
- 50 μm span
- **50 μm x 250 μm at 1 KHz**
- * 2ns $\rightarrow$ cross talk?! GCaMP6f/s

J. Wu ... M. Z. Lin, K. K. Tsia, and N. Ji, *Nat Meth* (2020).

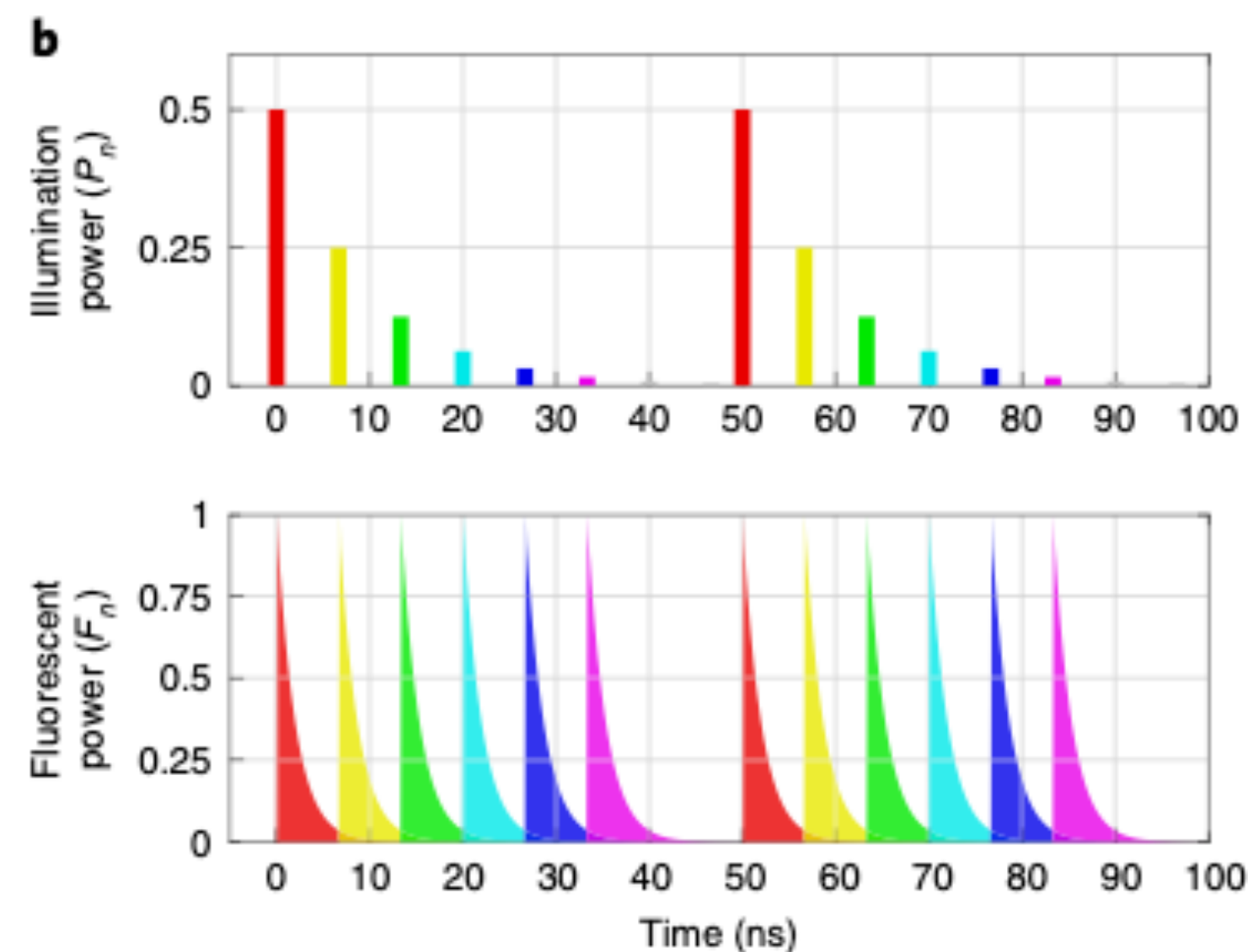
J.-L. Wu ... K. K. Tsia, *Light Sci Appl* 6, e16196 (2017).

Spatiotemporal MUX in Z (axial)

Reverberation Microscopy^a



t = 6.7 ns (~9% cross talk)

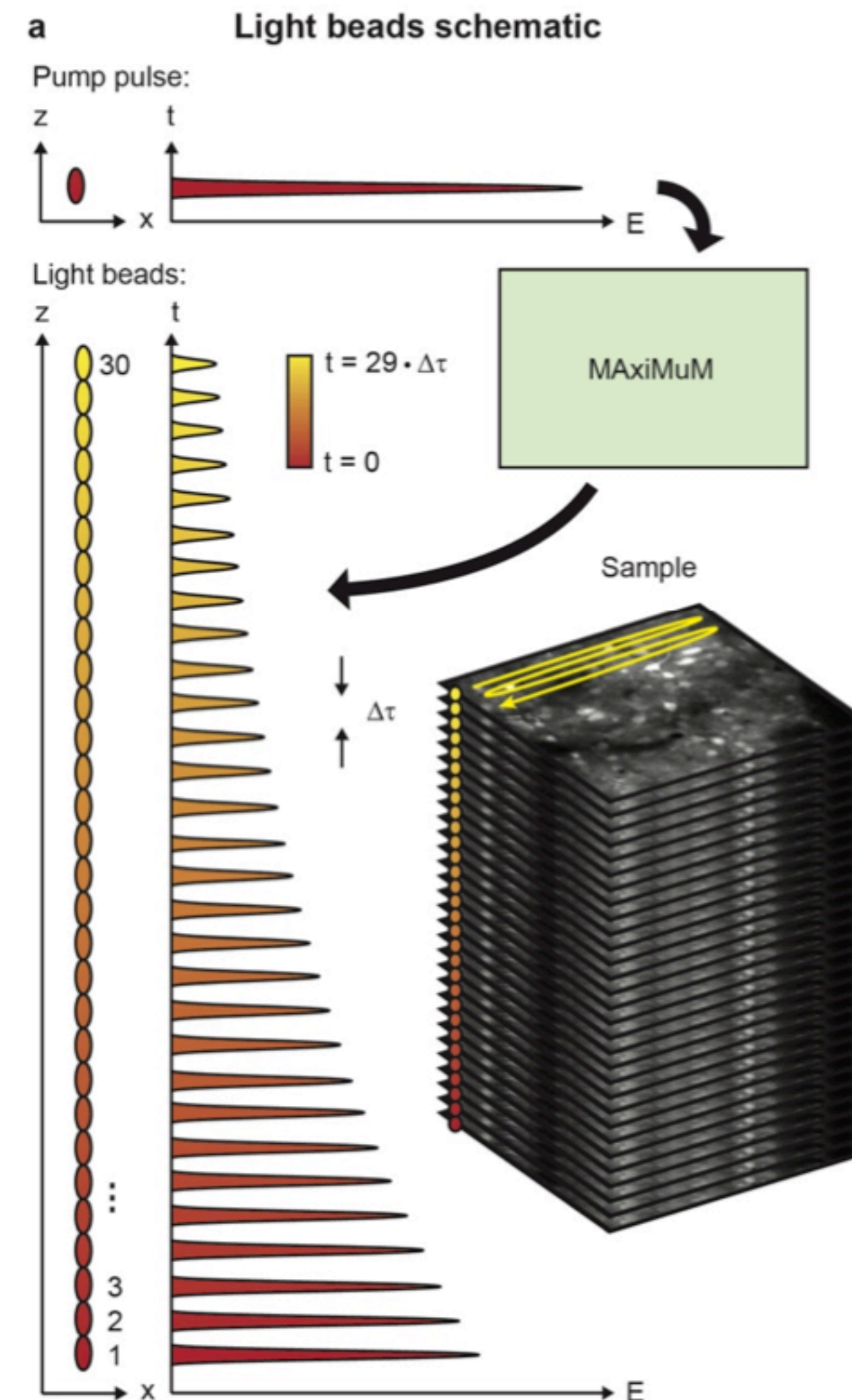
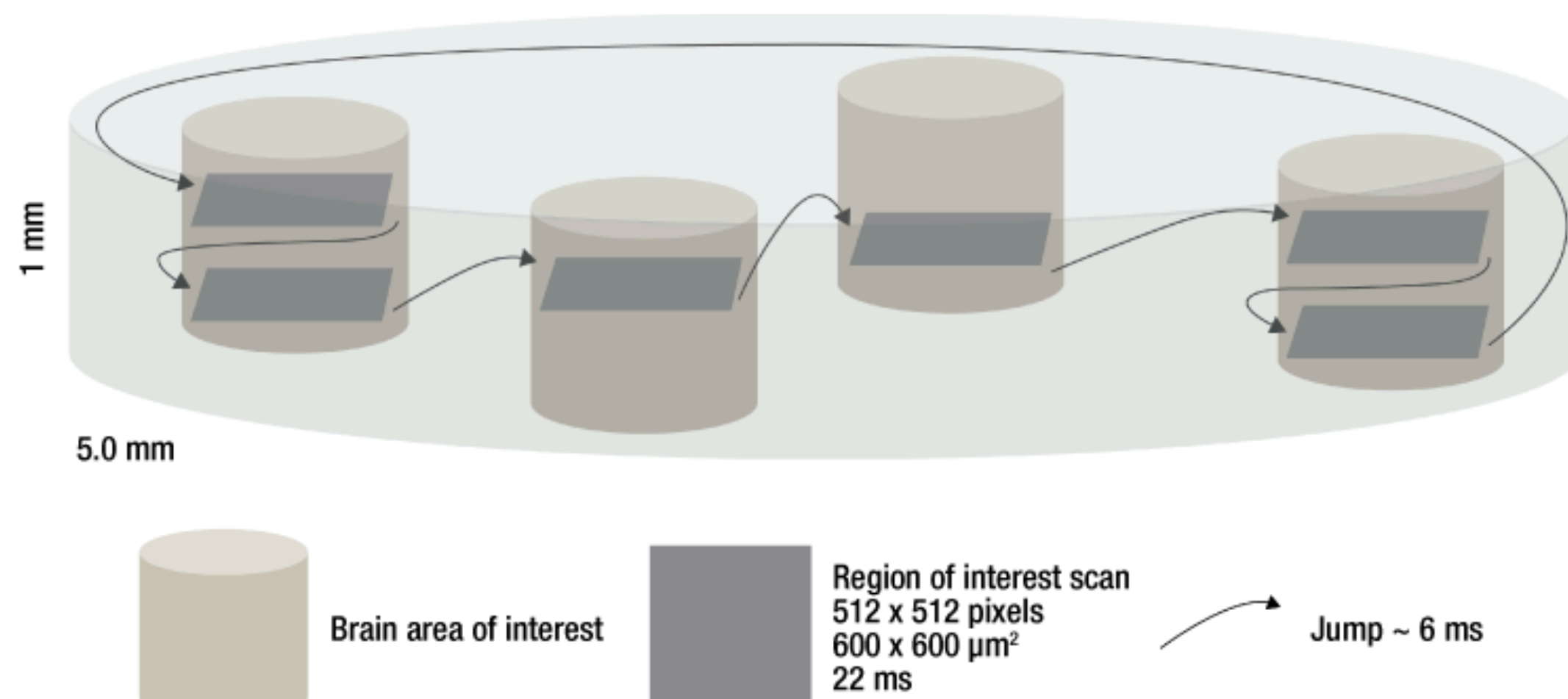


Power from loop: $P_i = T(1 - T)^i$

Fluorescence: $F \propto F_0 \exp(-D/l_s)$

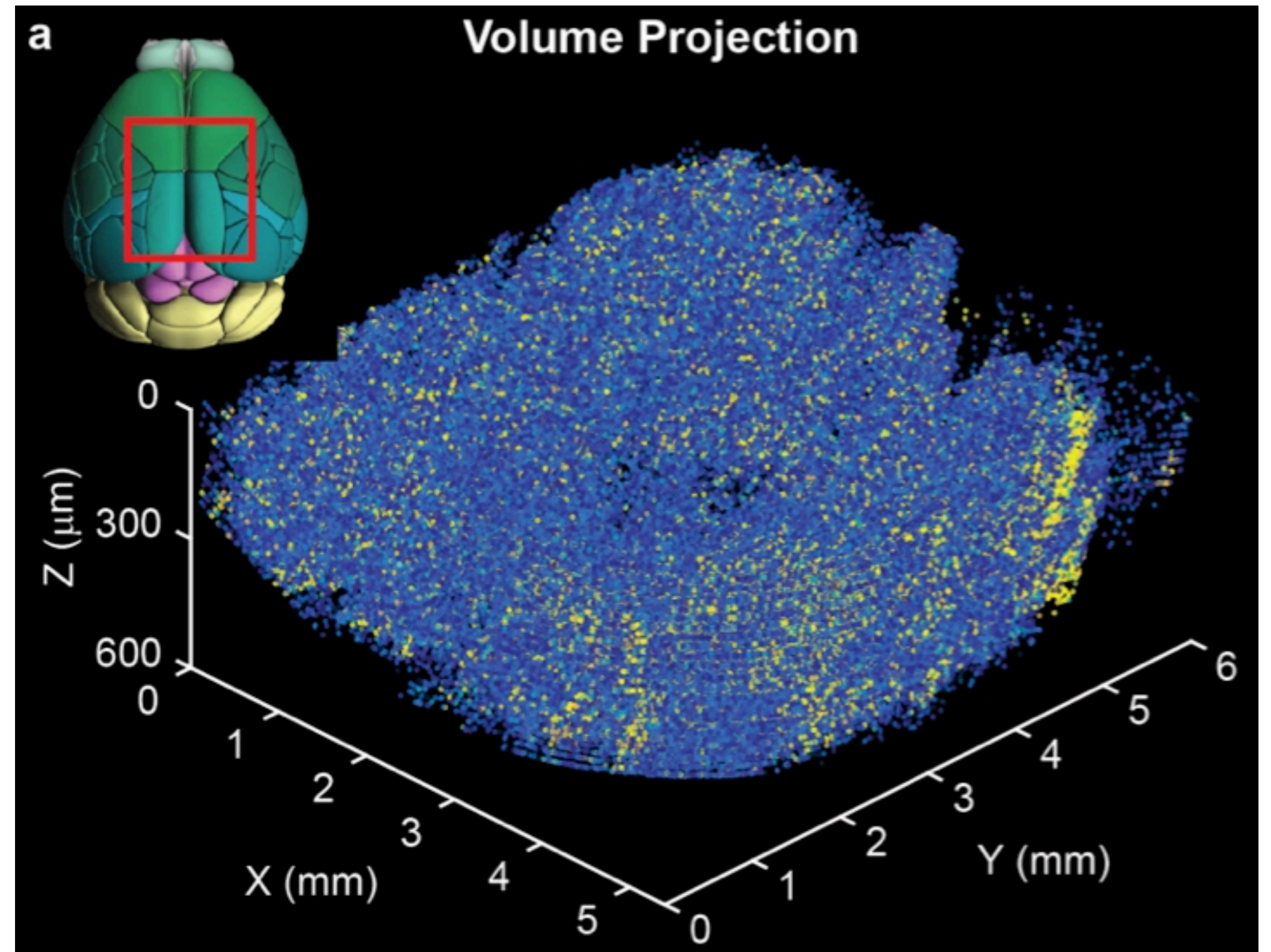
Light Bead Microscopy

- ‘Optimal excitation’:
 - Temporal separation is $\sim 7\text{ns}$
 - $\delta z = 16\ \mu\text{m}$ (30 beads = $500\ \mu\text{m}$)
- Laterally:
 - Laser rep rate to deliver 1 pulse every $5\ \mu\text{m}$ for $600\ \mu\text{m}$
- Coupled with the Thorlabs Mesoscope (Svoboda Lab) get to $5\ \text{mm}^2$ FOV



~1M Neurons!!

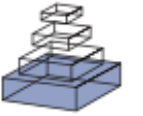
- 807,000 neurons in $\sim 5.4 \times 6 \times 0.5 \text{ mm}^3$ volume at 2.2 Hz
- 1 pulse / voxel
- Recording bilaterally from primary visual, primary somatosensory, posterior parietal and retrosplenial



“Stevensons Law”

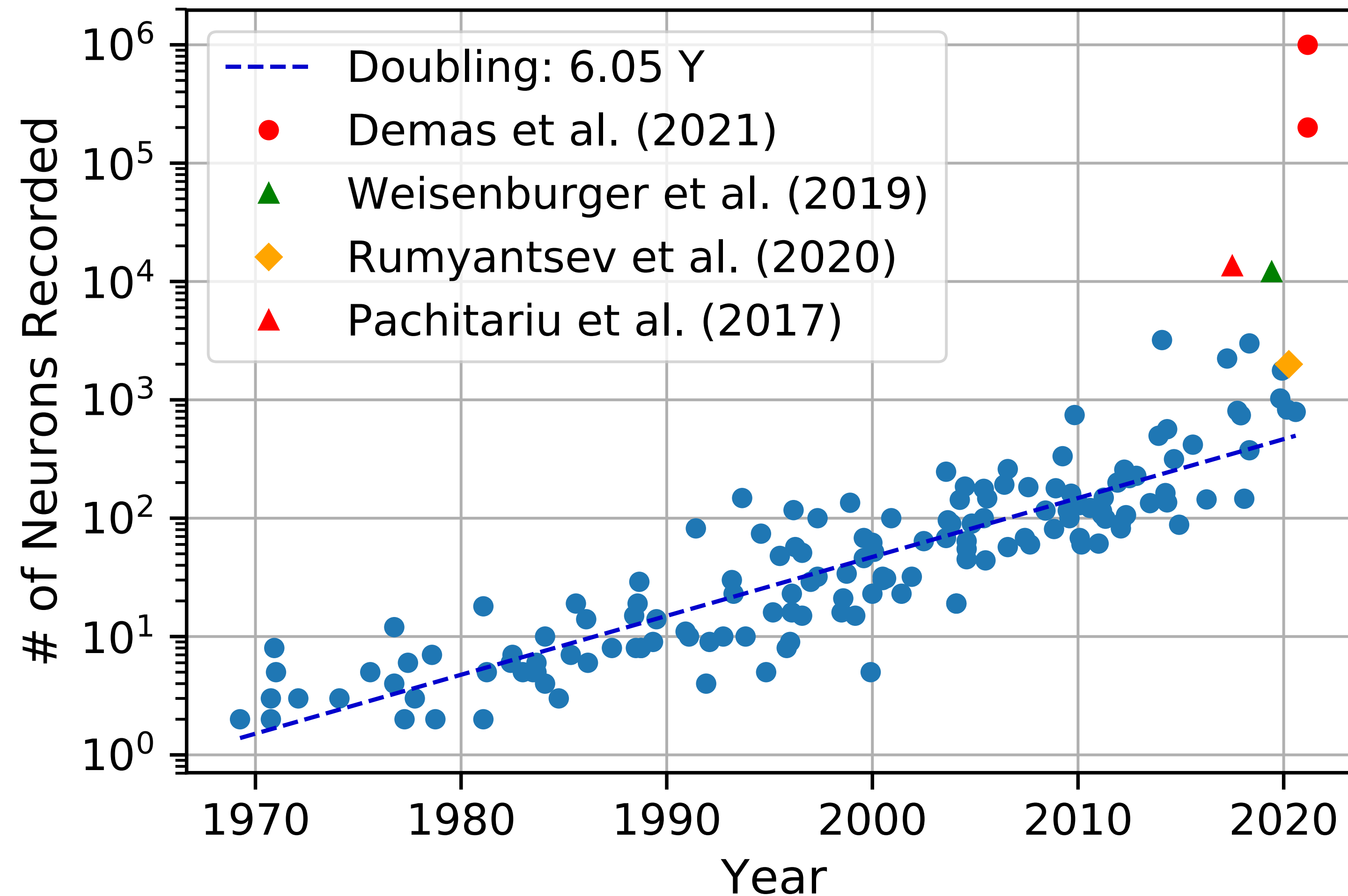
How advances in neural recording affect data analysis

Ian H Stevenson¹ & Konrad P Kording¹⁻³



Physical principles for scalable neural recording

Adam H. Marblestone^{1,2,*†}, Bradley M. Zamft^{3†}, Yael G. Maguire^{3,4}, Mikhail G. Shapiro⁵,
Thaddeus R. Cybulski⁶, Joshua I. Glaser⁶, Dario Amodei⁷, P. Benjamin Stranges³, Reza Kalhor³,
David A. Dalrymple^{1,8,9}, Dongjin Seo¹⁰, Elad Alon¹⁰, Michel M. Maharbiz¹⁰, Jose M. Carmena^{10,11},
Jan M. Rabaey¹⁰, Edward S. Boyden^{9,12‡}, George M. Church^{1,2,3‡} and Konrad P. Kording^{13,14‡}



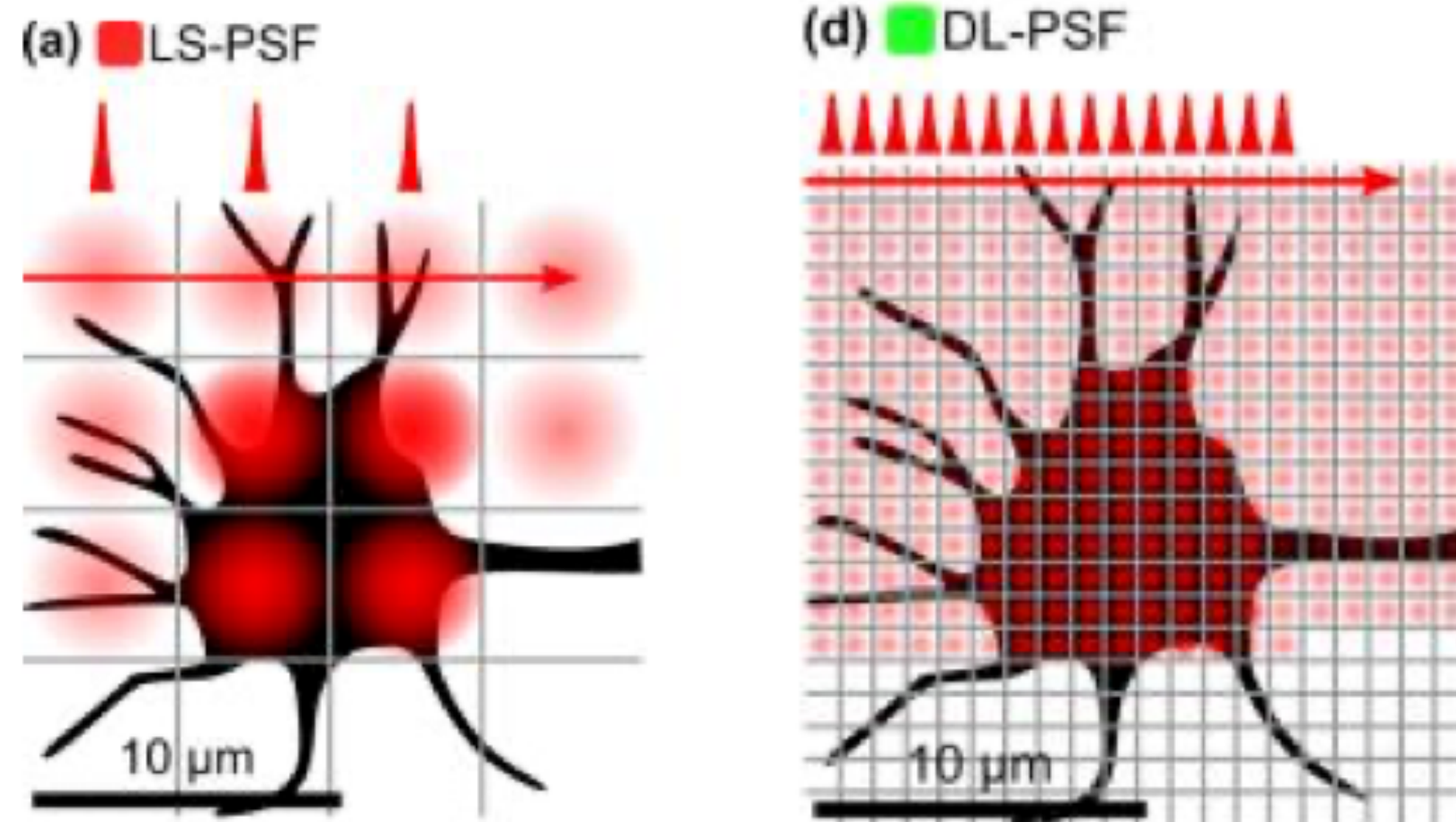
<https://stevenson.lab.uconn.edu/scaling/>

[†]S Herculano-Houzel et al., (2013)

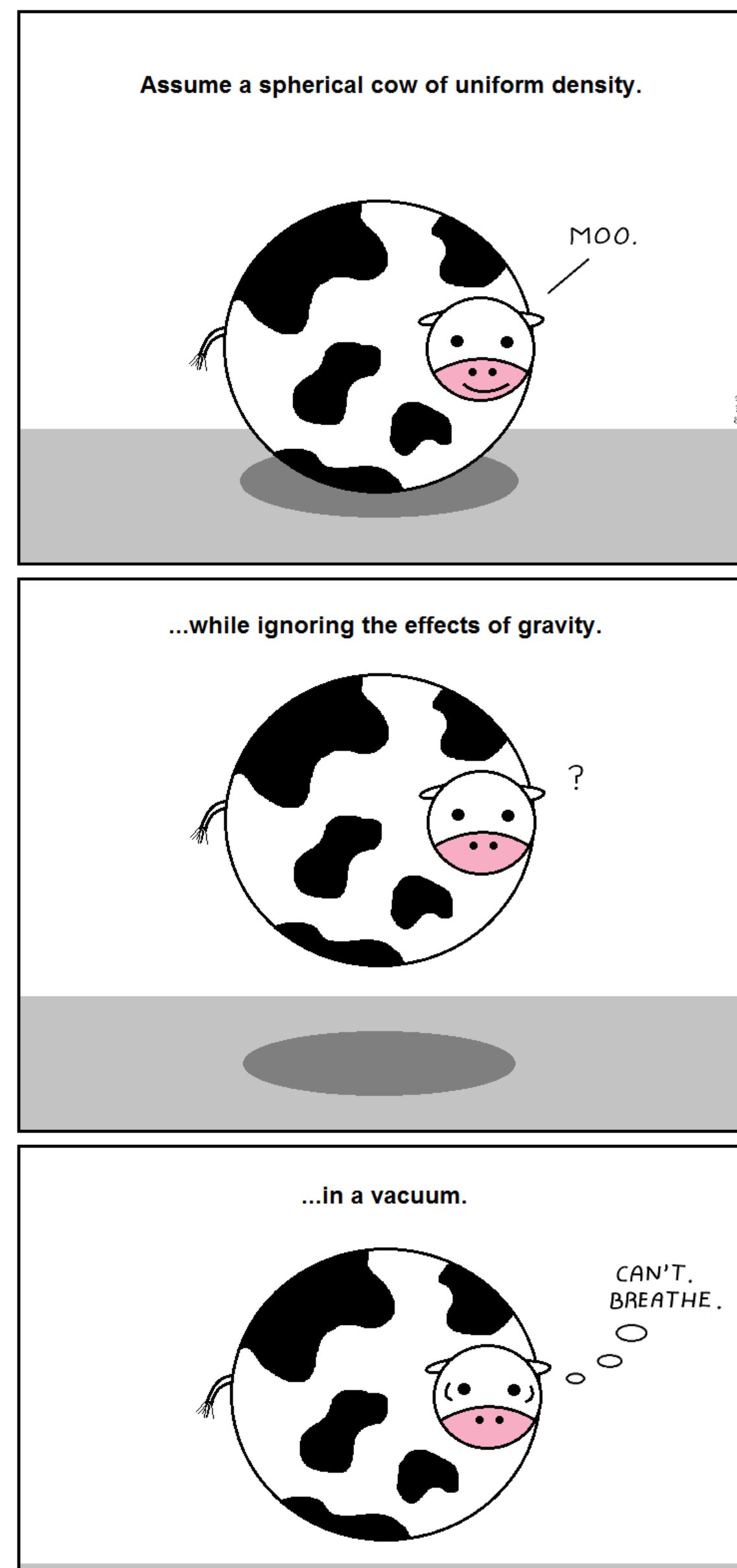
Trade-offs

Quantitative evaluation of two-photon calcium imaging modalities for high-speed volumetric calcium imaging in scattering brain tissue

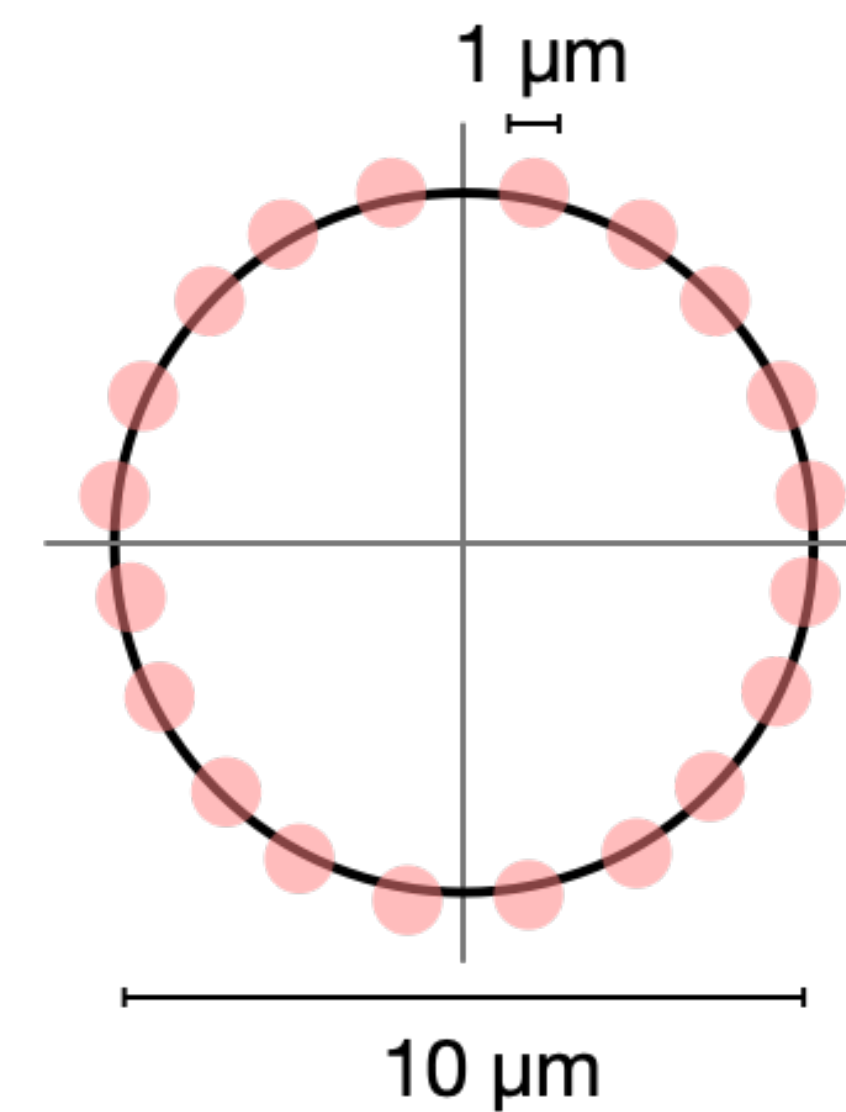
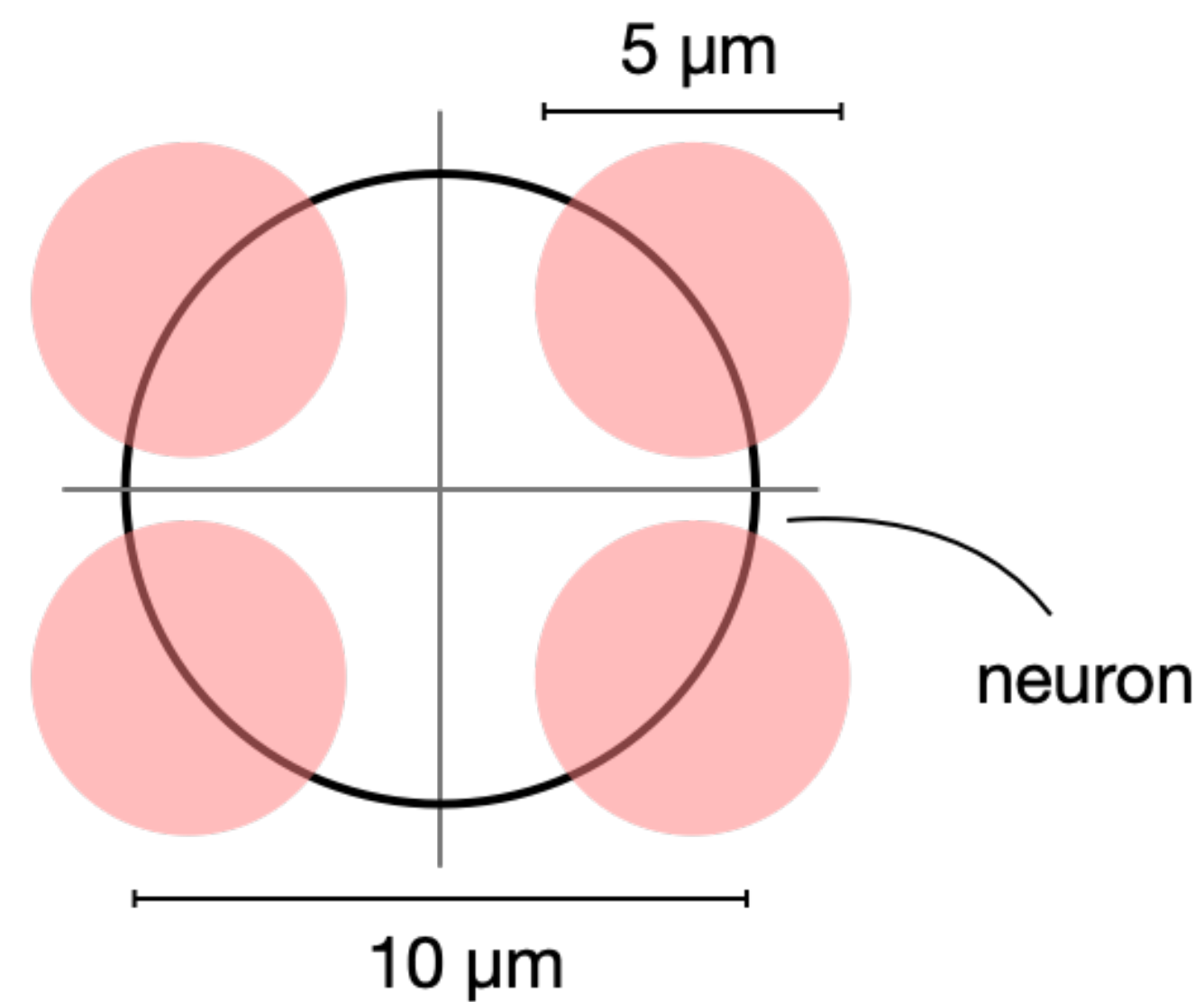
Siegfried Weisenburger^{1,2*}, Robert Prevedel^{2,3*} and Alipasha Vaziri^{1,2,†}



1. Less voxels → faster frame rate
2. For a given frame rate, less voxels → lower rep rate ∴ higher pulse energy
3. For a given intensity, more fluorophores are excited ∴ greater SNR (similarly for a fixed signal you can decrease the intensity)



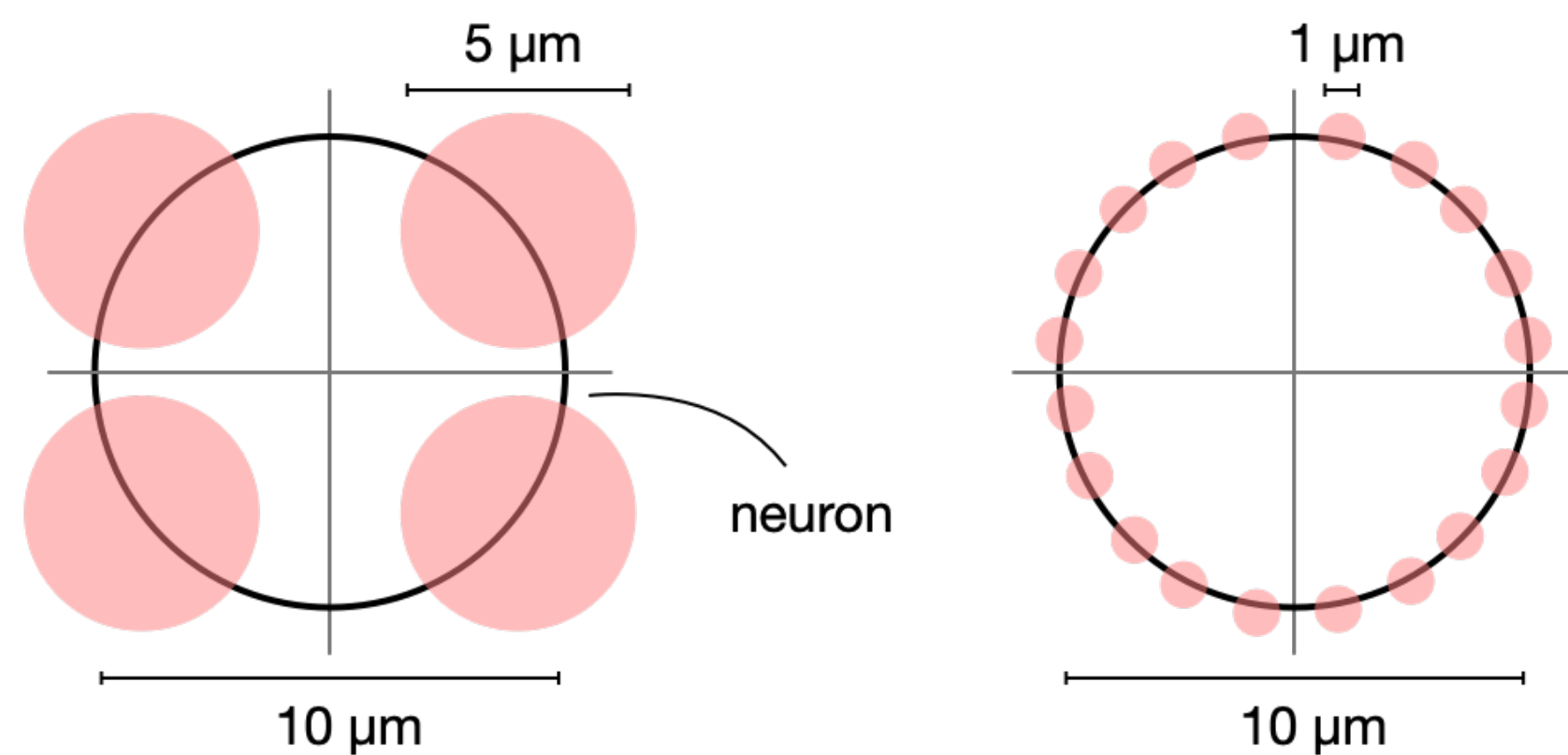
Abstruse Goose



$$N_a \sim C \times \frac{P_0^2}{f^2 \tau} \left(\frac{\lambda}{A} \right)^2 V$$

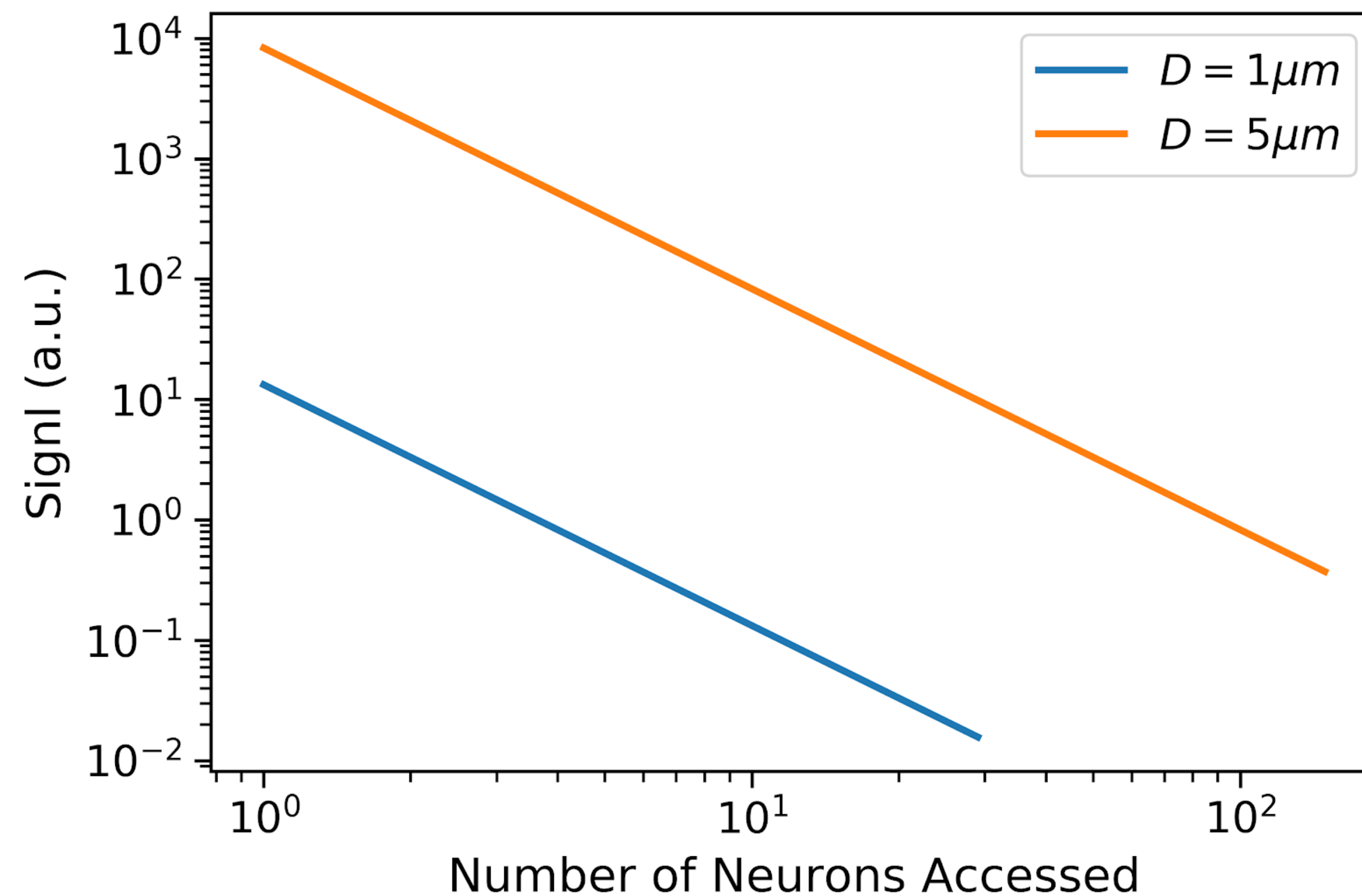
↑
|Pulse Energy|^2 ↑
Volume

C = constant
P₀ = average power
f = laser rep rate
τ = pulse width
A = illumination area
V = illuminated volume
containing fluorophores

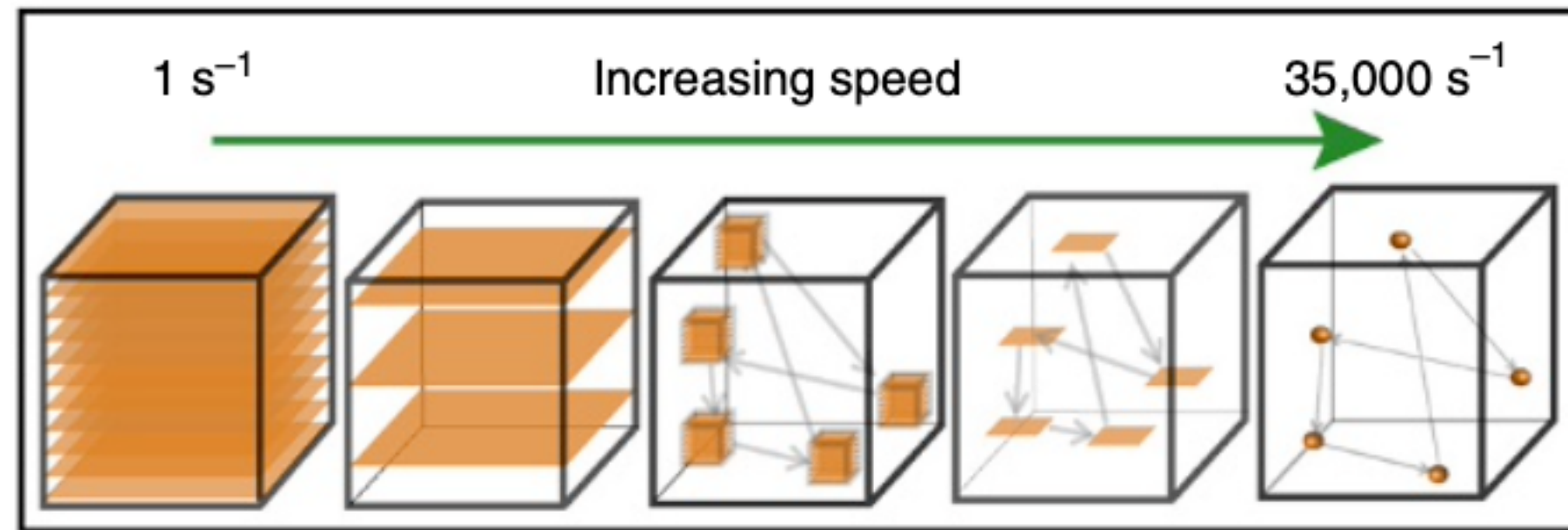


Assuming:

- Fixed energy density (E/A)
- Acquisition rate of 1 KHz
- 1 pulse / voxel
- Switching time of $1.5\ \mu\text{s}$

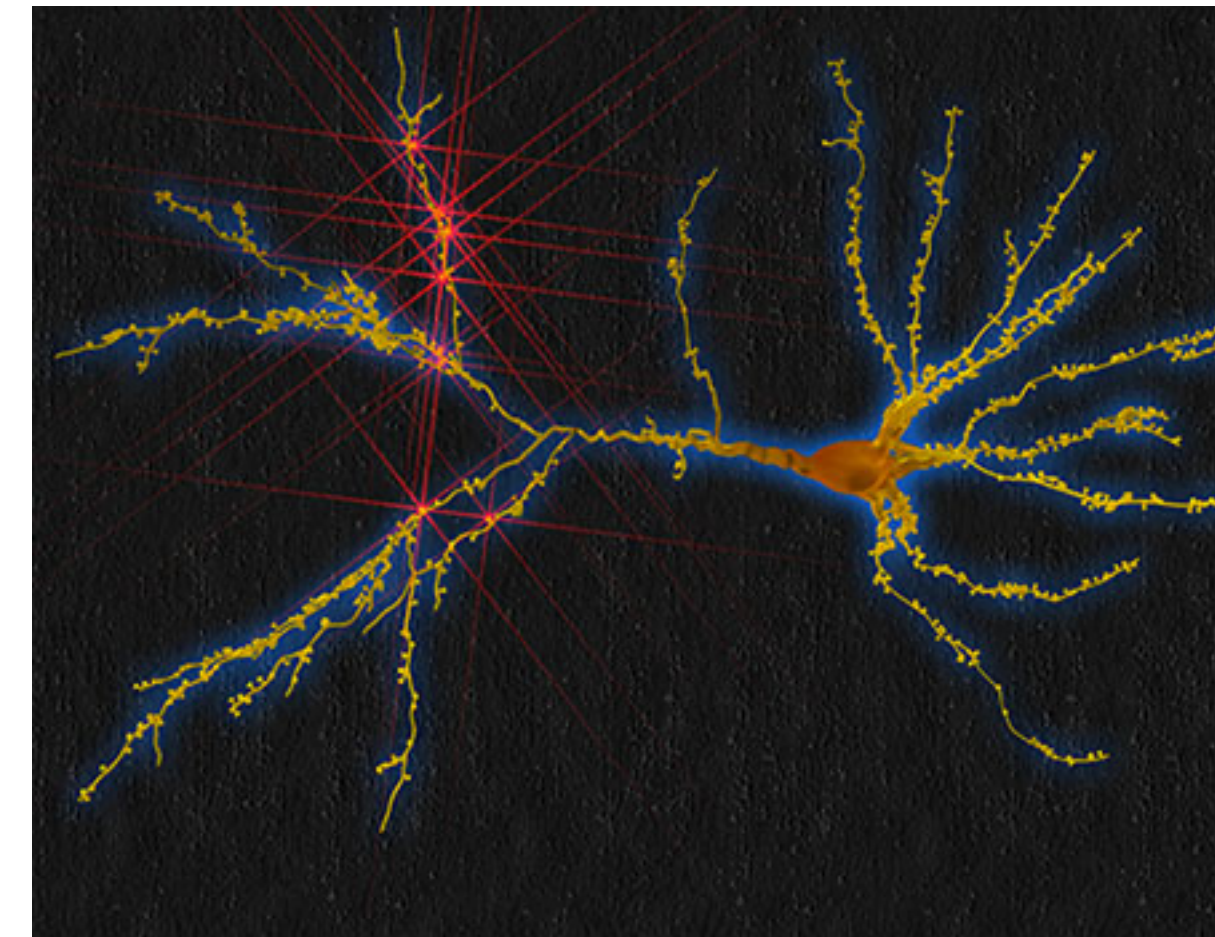


Random Access



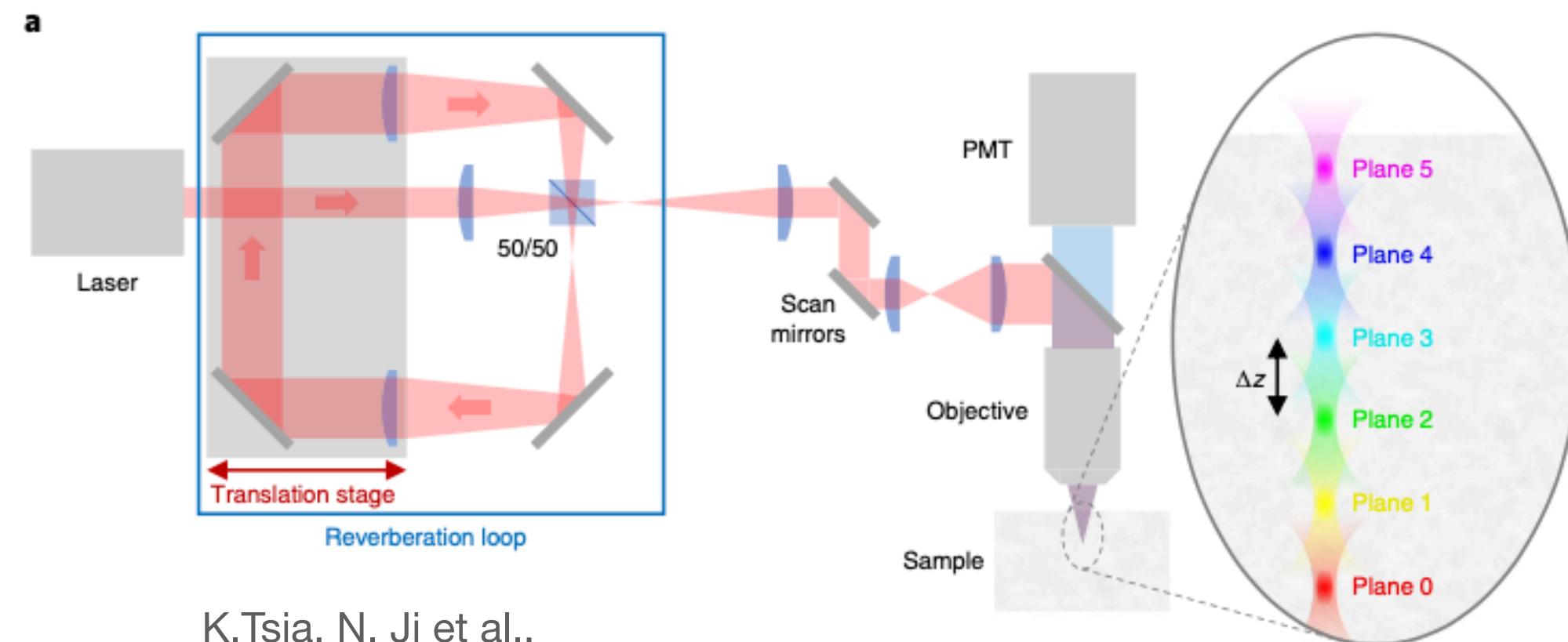
K. M. N. S. Nadella ... P. A. Kirkby, and R. A. Silver, *Nat Meth* (2016).

Computational



K. Podgorski

Multiplexing



K.Tsia, N. Ji et al.,

Thanks!

- New technologies
- Where we could do something
- Would love to chat about these ideas or any others